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Regulatory governance of the digital economy: Lessons from the financial services sector

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Abstract

This working paper examines whether lessons from financial sector regulation can inform the regulatory governance of the digital economy. Drawing on case studies from the Great Financial Crisis of 2008-2010 and the emergence of Digital Finance, the study identifies six challenges: information asymmetries between supervisory authorities and market participants; systemic risk recognition; regulatory adaptability; path dependencies arising from the timing of intervention; regulatory effectiveness; and international co-ordination. For each challenge, the paper derives lessons and assesses their relevance for the digital economy. The analysis reveals substantial structural parallels between the two domains, while identifying factors that amplify regulatory challenges in digital contexts – including the speed of technological change, the opacity of algorithmic systems, cross-sectoral business models, and accelerated path dependencies. The paper concludes that effective governance of the digital economy demands resource commitments commensurate with those required for post-crisis financial sector reform.

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Abbreviations and acronyms

ABSs	Asset-backed securities
AI	Artificial Intelligence
APIs	Application programming interfaces
BRIDGE	Better Regulation in the Digital aGE
CDOs	Collateralised debt obligations
FATF	Financial Action Task Force
FSA	Financial Services Authority
FSB	Financial Stability Board
FSF	Financial Stability Forum
MBS	Mortgage-backed securities
MiCAR	Markets in Crypto-assets Regulation
OTC	Over-the-counter
SVB	Silicon Valley Bank

Executive summary

Governments overseeing the digital transformation confront challenges that financial regulators faced – and often failed to master – during and after the Great Financial Crisis. The structural parallels are substantial and include a number of phenomena: cross-border operation that defies territorially organised regulatory authority; rapid innovation that outpaces legislative adaptation; systemic risk that propagates through networks; and information asymmetries between sophisticated service providers and regulators. This working paper argues that financial regulatory experience – encompassing both failures and successful adaptations – offers lessons directly applicable to the governance of the digital economy.

The study identifies six challenges to regulatory governance of the transition to the digital economy through case studies drawn from the Great Financial Crisis of 2008-2010 and the emergence of Digital Finance. For each, it derives lessons and assesses their relevance for governing the digital economy.

- **Information asymmetries.** Regulators cannot govern what they cannot see. Voluntary disclosure does not remedy this: market participants lack incentives to provide information that might invite intervention. The lesson is straightforward – where deemed necessary, mandatory reporting infrastructure *could* be established before activities achieve systemic scale, not after a crisis reveals the blind spot, or stronger incentives should be identified.
- **Systemic risk.** Risk can propagate through operational dependencies on concentrated infrastructure, behavioural patterns amplified by algorithmic systems, and the traditional channels such as bank run, and liquidity or collateral shortages. The digital economy introduces additional risk categories that established taxonomies do not capture. Regulation should incorporate processes for continuous reassessment of risk categories as technological conditions evolve.
- **Regulatory adaptability.** Static rules cannot govern dynamic systems. The case studies reveal a recurring pattern: reactive regulation arrives too late, prescriptive rules enable formal compliance while allowing market participants to circumvent regulatory substance, and crisis responses improvised under time pressure perform worse than pre-specified protocols. Effective regulation requires anticipatory capacity – the authority to intervene based on forward-looking risk assessment – and functional design that addresses economic substance rather than legal form or technological classification.
- **Path dependency.** Regulatory choices constrain future regulatory options. Delayed intervention allows market structures to entrench, and the costs of subsequent correction increase significantly. Arguments that regulation stifles innovation recur across contexts; they prevailed before the Great Financial Crisis and reappear in contemporary debates on digital economy governance. History suggests caution: the failure to regulate innovative markets has repeatedly produced private gains for market participants while generating large-scale social losses when conditions deteriorate.
- **Regulatory effectiveness.** Clear mandates, adequate resources, and sound methodologies are individually necessary but can be insufficient. Ambiguous mandates create accountability gaps; resource constraints produce supervisory failure regardless of formal powers; risk-based approaches risk rationalising inadequate capacity rather than optimising it (OECD, 2025[1]). Institutional design must guard against regulatory capture – a risk that intensifies where information asymmetries favour supervised entities over supervisors.

- **International co-ordination.** Regulatory costs fall on jurisdictions that enforce standards; risks can propagate from jurisdictions that do not. This asymmetry drives underinvestment in co-ordination and pressure toward a race to the bottom in regulatory standards driven by cross-border arbitrage. Voluntary co-operation fails under stress, when jurisdictions typically prioritise domestic interests. Binding protocols – specifying authority allocation, information-sharing obligations, and decision-making procedures – are more effective when established before crises materialise, not negotiated during them.

These six challenges intensify in the digital economy relative to financial markets, for reasons developed in the conclusion: technological change outpaces regulatory adaptation by a wider margin, algorithmic opacity creates information asymmetries of a kind not seen before in complex financial instruments, cross-sectoral business models fragment oversight, and network effects generate path dependencies within years rather than decades.

Lastly, effective governance of the digital economy demands resource commitments matching or exceeding those required for post-crisis financial sector reform: technical infrastructure for real-time monitoring, multidisciplinary expertise spanning technology, law, and economics, and compensation competitive with the technology sector. Failing to invest sufficiently in regulatory frameworks or infrastructure means carrying the risks arising from digital economy

1 Introduction

The digital transformation of the private and public sectors generates significant opportunities, contributing positively to growth while simultaneously addressing some of the world's most pressing challenges. Consequently, governments largely endorse increasing digitalization, generally taking a positive stance towards developments such as the platformisation of services, the use of AI in a wide range of applications, and the general datafication of the economy at large.[1]

At the same time, governments address the risks posed by the digital economy, which has the potential to threaten certain aspects of individual rights or individual welfare, or harm the proper functioning of markets, thereby making them less efficient, or endangering other societal values. State regulation is thus generally designed to foster and incentivise digital transformation, while simultaneously using specific mechanisms to mitigate the associated risks.

Legislators and regulators face challenges devising the appropriate regulatory response. Adapting the existing regulatory framework to optimally address the relevant regulatory challenges with a view to protecting society is a complex task, in a digital economy that evolves rapidly and continuously, and in ways which are difficult to predict (OECD, 2025[2]). The pace and transboundary nature of digital technologies require flexible and agile regulatory processes, moving beyond static "regulate-and-forget" approaches towards adaptive governance systems (OECD, 2025[2]; 2025[3]). Beyond regulation itself (the laws, orders, subordinate rules and administrative formalities that place requirements on enterprises and citizens), the concept of regulatory policy is broader and encompasses alternatives to regulation, such as voluntary standards. Regulatory governance maximises the influence of regulatory policy to deliver a positive impact on the economy and society and meet underlying public policy objectives. It is concerned with the design and implementation of regulation as well as ensuring compliance. It implies an integrated approach to the deployment of regulatory policies, tools and institutions (OECD, 2012[4]).

This report applies lessons from the financial services sector to the regulatory governance of the digital economy. In doing so, it is recognised that there is no individual "digital sector" but rather that the process of digitisation is a transformative force across the entire economy. The digital economy is viewed as a multi-dimensional ICT-driven segment of the economy, encompassing both the production of digital goods and services and the broader economic activities that are enabled or transformed by digital technologies. (OECD, 2024[5])

The Better Regulation in the Digital aGE (BRIDGE) initiative aims to provide support to governments taking relevant steps to further develop regulatory policy for the digital economy. It addresses knowledge gaps in regulatory practices and helps legislators and regulators in navigating the digital transformation. For this purpose, the initiative gathers insights on strategies employed in other regulatory areas that can be instrumental in shaping the relevant response. This analysis is underpinned by the OECD Going Digital Integrated Policy Framework 2026, which provides guidance on putting a whole-of-economy and society approach to digital policy making into practice.

[1] Platformisation refers to the phenomenon of large, scalable networks of users that facilitate access to a broader range of financial and non-financial products and services.

Opportunities and challenges flowing from the digital transformation

The digital transformation creates tension within regulatory governance: the same characteristics that generate economic and social benefits simultaneously create risks that may warrant intervention (OECD, 2025[2]). Policymakers must shape regulatory policy so that it mitigates harm without unduly constraining beneficial innovation – a balance complicated by uncertainty regarding how novel technologies and business models will evolve and interact with existing market structures. Several features of the digital transformation render this task particularly challenging.

The platformisation of services generates network effects whereby value increases with user participation,[2] creating powerful tendencies toward market concentration. While such concentration may yield efficiency gains, it simultaneously raises concerns regarding market power, consumer dependency, and the systemic relevance of some of the platforms whose disruption would affect multiple sectors. Regulatory frameworks must address these dynamics without predetermining market structure or impeding competitive entry. Further, the pervasive collection and analysis of personal data enable service personalisation and operational efficiency yet generates privacy risks and potential for exploitation of information asymmetries (OECD, 2025[6]; 2023[7]).

For example, cybersecurity and operational resilience concerns, as disruption to critical systems – whether through malicious attack or operational failure – may propagate widely through interconnected networks (OECD, 2025[6]). Furthermore, the deployment of AI in decision-making processes raises questions of accountability, transparency, and algorithmic bias that existing regulatory categories may inadequately address. Cross-sectoral business models compound these challenges (OECD, 2024[8]) (OECD, 2022[9]), as technologies and their applications fall within the purview of multiple policymakers and regulators, creating overlapping jurisdictions and potential regulatory gaps (OECD, 2025[2]). These risks require regulatory responses that account for both individual harm and systemic consequences.

The pace of technological change shortens the timeframes within which regulators must develop expertise regarding emerging harm and requires regulatory frameworks capable of continuous adaptation (OECD, 2023[7]; 2025[2]).

Learning from the regulation of financial markets

This study on lessons from the financial sector for regulatory governance of the digital economy aims to provide insights of relevance for addressing the challenges presented by digital transformation. It considers the regulation of the financial sector in response to two recent shifts of the market environment, i.e., the Great Financial Crisis 2008-2010 and the gradual shift of that industry to what is commonly called Digital Finance, roughly beginning around the year 2000 and gaining pace since about 2010.

The financial sector constitutes a particularly instructive source of regulatory experience for governments addressing challenges of regulating the digital economy. This is not coincidental: both domains share structural characteristics that generate analogous governance problems. Financial regulation has undergone intensive reform in recent decades that permits assessment of regulatory standard setting and enforcement across varying conditions.

Several features render the experience from regulating financial services directly transferable. First, both domains exhibit pronounced cross-border characteristics that challenge territorially organised regulatory authority. Financial institutions operate through complex corporate structures spanning multiple jurisdictions, exploiting regulatory divergences through arbitrage strategies; digital platforms similarly provide services globally from concentrated operational bases, rendering traditional territorial jurisdiction inadequate for effective oversight. Second, both domains feature rapid business model innovation that outpaces the adaptive capacity of regulation. Financial engineering produced instruments – from

[2] See Chapter 2 on the expanding mandates of communication regulators on broader digital areas, [Trends in access and connectivity: OECD Digital Economy Outlook 2024 (Volume 2) | OECD]().

securitisation structures to credit default swaps – that regulators struggled to comprehend and classify within existing frameworks; digital markets pose the same difficulty: platform intermediation models, algorithmic systems, and data-driven services likewise sit awkwardly within existing regulatory categories. Third, both domains present systemic risk characteristics whereby individual entity failures can propagate through interconnected networks to threaten broader stability. Financial contagion operates through balance sheet linkages, counterparty exposures, and confidence channels; digital infrastructure interdependence creates analogous dependencies whereby disruption to cloud services, payment systems, or communication platforms affects multiple sectors simultaneously. Fourth, cybersecurity and operational resilience are shared concerns. Financial institutions and digital service providers both face threats from malicious actors targeting critical infrastructure. Fifth, both domains raise consumer protection and data privacy challenges rooted in structural imbalances of bargaining power. Here, the information asymmetry between sophisticated service providers and individual users is what creates the scope for exploitation. Sixth, innovation in both sectors generates regulatory dilemmas between facilitating beneficial development and preventing harm accumulation, with competitiveness arguments frequently deployed to justify forbearance.

These parallels suggest that lessons derived from financial regulatory experience – encompassing both failures and successful adaptations – may be able to inform regulatory policy for the digital economy, while accounting for intensifying factors that may amplify governance challenges in digital contexts.

Methodology

This study employs a comparative case study methodology to derive transferable lessons from financial sector regulatory experience. The analytical approach proceeds in three stages.

The first stage identifies regulatory governance challenges that emerged prominently during two transformative episodes in financial markets: the Great Financial Crisis of 2008-2010 and the emergence of Digital Finance from approximately 2000 onwards. Rather than attempting comprehensive coverage of financial regulation, this study isolates six discrete challenges that proved particularly consequential for regulatory effectiveness: information asymmetries between supervisory authorities and market participants; systemic risk recognition and categorisation; regulatory adaptability to novel circumstances; path dependencies arising from the chosen timing of intervention; regulatory effectiveness including mandate specification and resource adequacy; and international co-ordination mechanisms. These challenges were selected because they recur across different regulatory contexts and possess general applicability beyond the financial sector.

These challenges are identified through case studies drawn from financial regulatory experience. Case studies are selected to illustrate specific regulatory dynamics rather than to provide exhaustive historical accounts. For each case study, the analysis identifies the regulatory framework in operation, the specific governance failure or success observed, and the mechanisms through which outcomes materialised. The case study evidence draws upon official reports, academic literature, and regulatory documentation, with particular attention to post-crisis reviews that benefit from hindsight regarding regulatory design choices.

The second stage derives lessons from the case study analysis and assesses their relevance for the governance of the digital economy. For each challenge, the study examines how analogous phenomena manifest in digital economy contexts, identifies factors that may amplify or attenuate the challenge, and formulates guidance for regulatory design. This comparative assessment acknowledges both the transferability of financial regulatory experience and the distinctive features of digital markets that may require adapted approaches.

In the third stage, the study formulates lessons that may be useful in understanding the different dimensions of regulating the digital economy. The study does not advocate for direct transposition of financial regulatory instruments to digital contexts. Rather, it employs financial regulation as a laboratory that has generated extensive evidence regarding regulatory policy design choices, implementation challenges, and unintended consequences. The objective is to enable governments to benefit from this accumulated experience when construing a regulatory framework for the digital economy, avoiding repetition of documented failures while adapting successful approaches to novel circumstances.

The experience of responding to the Great Financial Crisis

The 2008-2010 "Great Financial Crisis'' exposed fundamental weaknesses in regulatory governance and transformed understanding of financial system risk. The crisis provides instructive evidence regarding regulatory design choices, implementation challenges, and institutional failures from which lessons applicable to other contexts can be derived.

The crisis originated in the United States mortgage market, where loans extended to borrowers with weak credit profiles were repackaged through securitisation into structured products and sold as highly rated investments. These instruments were further repackaged into successive layers of derivative products, creating opacity regarding underlying risk exposures. When interest rates rose and borrowers defaulted, the value of these instruments collapsed, revealing risk concentrations that had been obscured by financial engineering. The resulting loss of confidence precipitated a liquidity crisis as interbank lending froze, leading to widespread institutional failures across multiple jurisdictions.

This chain reaction occurred within a specific market context. Macroeconomic imbalances and persistently low interest rates had fuelled excessive borrowing, while financial institutions operated with leverage ratios that left minimal capacity to absorb losses. Risk measurement methodologies proved inadequate, and incentive structures emphasising short-term performance metrics encouraged risk accumulation.

The regulatory and supervisory frameworks in place were inadequate. Regulatory authorities were understaffed and operated under "light-touch" assumptions that trusted in market self-correction. The interconnectedness of global financial institutions was poorly understood, and supervisory setup fragmented oversight responsibilities across institutional structures and jurisdictional borders, constraining comprehensive assessment of system-wide vulnerabilities. This fragmentation provided banks with arbitrage opportunities, including exploitation of capital requirements that enabled concealment of exposures, and utilisation of shadow banking structures operating outside the prudential framework.

Post-crisis regulatory reform addressed these concerns through enhanced micro-prudential standards and the introduction of macro-prudential oversight mechanisms designed to monitor systemic risk. International regulatory co-ordination was substantially strengthened to better monitor risks that arise and propagate across borders. However, while governments demonstrated initial consensus under the immediate influence of the crisis, this commitment has subsequently diminished alongside efforts to address subdued economic performance in recent years, as demonstrated by recent deregulatory initiatives that partly roll back measures adopted post-crisis.

Relevant details are provided in the context of the case studies.

The emergence of Digital Finance

Since around 2015, the term Digital Finance (or "FinTech") has been used to refer to technology-enabled provision of financial services. The phenomenon encompasses several distinct but partially interrelated market developments.

The most visible development concerns structural change in how financial services are provided. Beginning with online payment options around 2000 and accelerating with smartphone technology from approximately 2010, financial services have migrated to digital channels. Online-only providers initially concentrated on specific services, such as payment services, before aggregating offerings to resemble traditional banks. This development relies, amongst other elements, upon the mobile device environment, application programming interfaces (APIs) and cloud technology. These technological achievements allow the transformation of the entire value chain and enable significant efficiency gains.

First, financial services are increasingly provided through platforms – large, scalable networks facilitating access to financial and non-financial products. Network effects, whereby value increases with user participation, create tendencies toward market concentration. The so-called BigTech companies dominate this environment as providers of platforms and cloud infrastructure, even where they do not directly offer financial services to customers. BigTech refers to platform-based large technology companies with extensive customer networks and conglomerate business models across multiple markets (OECD, 2025[10]).

Second, datafication refers to the application of analytical tools to comprehensive data on individuals, enabling financial service providers to assess creditworthiness, target offerings, and reduce compliance costs. Platforms and data aggregators possess access to extensive customer data that financial service providers can leverage, diminishing the competitive advantage of established customer relationships and enabling unbundling of financial services. Simultaneously, the provision of digital financial services generates valuable proprietary data that, when shared with platforms, enables targeting across diverse commercial purposes.

Third, artificial intelligence is deployed across customer-facing functions in the entire financial industry, wholesale market analysis, and back-office operations. Applications include services providing algorithmic investment advice, trading systems processing structured and unstructured data to predict market movements, and automated credit scoring, risk management, and regulatory compliance functions.

Fourth, recently, crypto-assets have emerged that resemble financial products, including "stablecoins", which are purporting to maintain a stable value and promise to be redeemable against money. These new products raise regulatory classification questions: where crypto-assets functionally correspond to regulated financial products, existing frameworks apply; where they do not, the question arises whether dedicated regulatory regimes are required.

Crystallising six challenges to the regulatory framework

This study identifies six challenges to regulatory governance that emerged prominently during the Great Financial Crisis and in the context of the emergence Digital Finance. While these transformative episodes differed substantially in character – crisis-driven in the former case, opportunity-driven in the latter – they generated analogous governance problems from which transferable lessons can be derived for the regulatory response to other emerging technologies under the digital transformation.

The six challenges are: i) asymmetry of information between market participants and regulators; ii) recognition and categorisation of systemic risk; iii) adaptability of regulation to novel circumstances; iv) path dependencies arising from regulatory timing and intensity; v) regulatory effectiveness, including mandate specification and resource adequacy; and vi) international regulatory alignment and supervisory co-ordination.

Regulatory policy challenges confronting the governance of the digital economy substantively replicate those exposed by financial crisis experience: information asymmetries favouring private actors over supervisory authorities; regulatory lag or enforcement lag enabling risk concentration in under-regulated market segments; arguments invoking a binary choice between facilitating innovation and protective regulation, where the reality is more nuanced; and cross-border operation enabling regulatory arbitrage. However, several factors amplify these challenges in digital contexts beyond those characterising financial regulation. The speed of technological evolution often exceeds the capacity of regulation to adapt, rendering regulation obsolete within short timeframes. The opacity of algorithms generates information asymmetries qualitatively distinct from those known so far, for instance those characterising complex financial instruments. Platforms and cross-sectoral business models are addressed in different areas of regulation and by different regulators. Network effects and data accumulation generate path dependencies within years rather than the decades over which analogous phenomena developed in financial markets. Effective regulatory governance accordingly demands resource commitments and institutional adaptation commensurate with — and potentially exceeding — those required for post-crisis financial sector reform.

The study examines each challenge through case studies drawn from financial regulatory experience (Sections 2–7), assessing their relevance for governance of the digital economy and formulating lessons for regulatory design. The conclusion (Section 8) synthesises these findings into guidance for governments and regulators addressing the ongoing digital transformation.

2 Asymmetry of information between market participants and regulators

Effective regulatory oversight depends upon authorities possessing adequate information regarding regulated entities' activities, risk exposures, and interconnections. Where market participants possess superior knowledge in this regard, the regulatory framework becomes vulnerable to both inadvertent blind spots and strategic exploitation by market participants. The following analysis considers how information asymmetries undermined financial regulation preceding the 2008 crisis and how they persist in novel segments of the financial services markets, deriving lessons applicable to regulatory governance of the digital economy.

Information asymmetries arise through the combination of regulatory perimeter gaps and absence of mandatory disclosure infrastructure. In a context where market participants undertake significant activities outside the established reporting framework, financial supervisors possess neither real-time data nor comprehensive understanding of aggregate exposures and interconnections. The opacity enables risk accumulation outside the perimeter visible to the supervisor, with systemic significance becoming apparent only during crisis episodes.

Both case studies discussed below show that voluntary disclosure proved generally inadequate for regulatory purposes. Market participants lacked incentives to provide comprehensive information absent relevant legal obligation, particularly where disclosure might reveal insights inviting supervisory intervention. In either context, effective access to information would have required a mandatory reporting framework, using standardised data formats and taxonomy as well as a suitable form of centralised repository.

Information asymmetries constitute a traditional challenge in market design (Akerlof, 1970[11]). Where regulators possess generally inferior information regarding risk exposures, business models, or market interconnections, the regulatory framework encounters three distinct difficulties. First, regulators cannot identify accumulating risk until it materialises in observable adverse outcomes, that is, *ex ante* intervention is not possible. Second, regulatory requirements formulated without comprehensive market intelligence may prove ineffective or generate unintended consequences through misalignment with the reality of the market and relevant market practices. Third, sophisticated market participants exploit information advantages to engage in regulatory arbitrage, for example, through structuring activities to achieve formal compliance while circumventing regulatory purposes ("box-ticking") (Sunstein, 2011[12]).

The severity of these asymmetries varies according to market segment and structure, and regulatory setup. Opaque products, complex corporate structures, and rapid innovation aggravate the situation. Where regulated entities control information, strategic behaviour widens asymmetries beyond those that occur already as a consequence of an expertise gap between supervisor and supervisee.

Case study 1: Over-the-counter (OTC) derivatives reporting

The OTC derivatives market prior to the Great Financial Crisis illustrates how bilateral market structures generate information asymmetries that impede effective regulatory oversight.

Prior to the Great Financial Crisis, the OTC derivatives market had grown to a notional value exceeding USD 600 trillion by 2008. This largely unregulated market was characterised by direct transactions between parties. During the Great Financial Crisis, OTC derivatives played a critical role in amplifying and spreading risk throughout the financial system. The collapse of Lehman Brothers and AIG's near-failure were closely linked to their extensive OTC derivatives exposures.

Regulators lacked comprehensive data on individual transactions and aggregate volumes, real-time information on market-wide exposures, detailed insights into inter-institutional connections, and accurate assessments of counterparty risks in relation to OTC derivatives (Gorton and Metrick, 2012[13]). This information deficit severely hampered their ability to identify and mitigate the significant systemic risk inherent in the relevant market segment.

Enhanced information availability could have substantially aided both pre-crisis risk management and crisis response. Pre-crisis, it could have facilitated the identification of exposure concentrations, enabling more suitable regulation, such as targeted capital surcharges or exposure limits for systemically important institutions. For instance, information on the magnitude at which credit default swaps were used in the securitisation market could have enabled regulators to address the concentration of risk in AIG group entities before they failed.

During the crisis, comprehensive data on derivatives exposures could have expedited the resolution process for failing institutions by providing a clearer picture of their liabilities. Real-time information on market-wide exposures could have helped regulators more accurately assess the potential impact of allowing a major institution to fail.

In response to the crisis experience, the G20's 2009 reform agenda demanded comprehensive changes to the OTC derivatives market, with a significant focus on reporting obligations. An increasing number of categories of OTC derivatives transactions must since be reported to trade repositories. Reportable data includes transaction details, product type, counterparty information, and valuation and collateral data. The Financial Stability Board, an international body established in the wake of the Financial Crisis, regularly assesses the implementation progress and effectiveness of these reforms across jurisdictions.

Gatekeepers – including stock exchanges, auditors, credit rating agencies, lawyers – theoretically mitigate information deficits by translating specialised operational knowledge into standardised disclosures accessible to regulators and market participants. However, conflicts of interest and reputational dependencies may compromise gatekeeper effectiveness, transforming these intermediaries from information producers into facilitators of opacity. Famously, credit rating agencies assigned favourable ratings to complex securitised products without adequate due diligence, as a result of being conflicted (Turner, 2009, pp. 77-78[14]).

Case study 2: Disclosure and reporting of "stablecoin" reserves

The stablecoin market is predominantly led by Tether ("USDT") and Circle ("USDC"), with market capitalisations of approximately USD 186 billion and USD 74 billion, respectively (as of January 2026). Stablecoins serve as trading pairs on cryptocurrency exchanges, collateral in crypto-asset lending operations or means of exchange in various contexts of the crypto-asset market. They are also on the cusp of integrating with traditional payment systems. The principal categories of stablecoin users encompass crypto-asset exchanges and other crypto-asset service providers, for which stablecoins serve as the dominant base trading pair and settlement currency, as well as over-the-counter (OTC) trading desks, hedge funds and other institutional investors with digital asset allocations, family offices and high-net-worth individuals. Increasingly, retail consumers in emerging markets rely on dollar-pegged stablecoins as a store of value, remittance instrument, and medium of exchange in economies characterised by high inflation, currency instability, or restricted banking access (FATF, 2026, pp. 7-9[15]); (PwC/AIMA, 2025[16]); (PwC/AIMA, 2025, pp. 14-16[16]); (Aki Yokoyama et al., 2025, p. 1[17]). Stablecoins have simultaneously become the instrument of choice for illicit finance: the FATF reported that stablecoins accounted for 84% of illicit virtual asset transaction volume in 2025, frequently involving unhosted wallets and complex laundering techniques (FATF, 2026, p. 5[15]), citing Chainalysis). Illicit use is dominated by sanctions evasion by state actors, notably Russia, Iran, and the DPRK, which accounted for 86% of illicit crypto-asset flows in 2025, alongside transnational organised crime networks engaged in cyber fraud, money laundering, and underground banking, particularly in East and Southeast Asia (TRM Labs, 2025[18]); (UNODC, 2024[19]). In absolute terms, TRM Labs estimated that illicit entities received USD 141 billion in stablecoins in 2025, the highest level observed in five years, though this represented less than 0.5% of total stablecoin transaction volume of at least USD 35 trillion (TRM Labs, 2025[18]).

The key areas of information asymmetry encompass the composition and quality of reserve assets backing stablecoins, the extent to which reported market capitalisation accurately reflects issuers' ability to honour redemption obligations at par, the profile of users and relevant types of usage, and the interconnectedness of the stablecoin market with the regulated financial system via market participants active in both (IMF/FSB, 2024, p. 3[20]); (FSB, 2025[21]). These challenges stem in large part from the absence or inadequacy of reporting obligations: for unregulated stablecoins such as USDT, disclosure remains voluntary and limited to periodic issuer-commissioned attestations rather than independent audits; for regulated stablecoins, emerging frameworks, notably MiCAR in the EU and the GENIUS Act in the United States, are beginning to impose reserve segregation, disclosure, and supervisory reporting requirements, but implementation remains uneven, with only five jurisdictions having finalised dedicated stablecoin frameworks as of August 2025 (FSB, 2025[21]); (FATF, 2026, p. 6[15]). There is consequently no suitable infrastructure in most jurisdictions to collect and analyse the data necessary for effective supervisory oversight. The Financial Stability Board's recommendations on stablecoins serve as a benchmark for jurisdictional regulatory responses, though implementation diverges substantially. The European Union's Markets in Crypto-assets Regulation (MiCAR) establishes comprehensive requirements for stablecoins marketed within the EU, including mandatory disclosure of reserve composition and management, reporting on issuance and redemption, and stress testing obligations. MiCAR restricts issuance of significant stablecoins to supervised financial institutions and mandates that 60% of reserves be held as deposits with credit institutions. Similarly, the United States GENIUS Act, enacted in 2025, establishes a framework permitting both bank and approved nonbank issuers through dual federal state licensing pathways, requiring 100% reserve backing using specified liquid assets and mandates monthly public disclosure of reserve composition.

Addressing asymmetric information between market participants and regulators, both inadvertent information deficits and the danger of strategically withheld information need to be considered. At the core of a suitable regulatory regime are reporting obligations. Their effectiveness depends on the correct specification of the scope and the frequency of reporting.

Supervisory authorities seek to allocate scarce resources efficiently ('risk-based supervision'); however, this presupposes that authorities possess sufficient information to assess risks accurately. This condition would not be met where material exposures remain unknown (Black, 2005[22]). Therefore, a regulatory framework is unlikely to function effectively if it relies exclusively on disclosed information without independent verification capacity. In such a scenario, reporting obligations may generate compliance formalism that does not reflect the substantive market reality and, hence, obfuscates the picture instead of creating transparency.

Relevance for regulatory governance of the digital economy

The preceding case studies reveal common patterns while exhibiting significant differences that inform the application of regulatory lessons to digital transformation contexts. This subsection first synthesises the comparative findings from the OTC derivatives and stablecoin analyses, then assesses their relevance for regulatory governance of the broader digital economy.

Comparative assessment

Both case studies confirm the pattern set out at the opening of this section: information asymmetries arise through the combination of regulatory perimeter gaps and the absence of mandatory disclosure infrastructure, and voluntary disclosure proves inadequate where market participants facing competitive pressure or regulatory scrutiny lack incentives to disclose information that might invite intervention or place them at a disadvantage relative to less transparent competitors. The episodes are different in important elements. OTC derivatives trading occurred primarily amongst sophisticated financial institutions subject to prudential supervision in their core banking activities, though derivatives exposures themselves evaded comprehensive oversight. Authorities theoretically possessed institutional relationships enabling information requests, even where legal frameworks did not mandate systematic disclosure. Stablecoins, conversely, often operate through entities that fall outside of the scope of regulatory perimeters, eliminating even informal channels for information access.

Temporal and cross-border dimensions also differ significantly. OTC derivatives markets developed over decades, affording regulators extended periods to observe growth and assess risks, though institutional inertia prevented timely response. Stablecoins achieved systemic scale within very few years, shortening the time available for regulatory deliberation. While OTC derivatives involved globally active institutions transacting through established financial infrastructure spanning multiple jurisdictions, stablecoins operate through digital networks permitting instantaneous global distribution unconstrained by physical presence or traditional intermediation. Jurisdictional differences in reporting requirements – where they exist – and challenges in information sharing between regulators, create additional blind spots that cross-jurisdictional co-operation arrangements have not adequately addressed.

On the surface, these information asymmetries appear to stem from the absence of adequate regulation, as a consequence of which information on sophisticated financial products remains unavailable to supervisory authorities. A visible gap in expertise and resources between regulators and private sector participants worsens the problem. In the example of the Great Financial Crisis, the failure to regulate innovative markets resulted in significant private gains for market participants while generating large-scale social losses. On the eve of the crisis, financial sector profits constituted 27% of all corporate profits in the United States, up from 15% in 1980 (Financial Crisis Inquiry Commission, 2011, p. xvi[23]). When the crisis materialised, US households lost approximately USD 19.2 trillion in wealth and the number of jobs decreased by 8.8 million (US Department of the Treasury, 2012[24]). Inflation-adjusted household wealth declined by 26% from its 2007 peak to early 2009, with younger and less wealthy families suffering disproportionately larger losses (Emmons and Noeth, 2012[25]). The financial benefits of unregulated activities thus accrued to a relatively small number of institutions, while taxpayers, small and medium-sized enterprises, and vulnerable economies bore the consequences. A similar dynamic may be unfolding in Digital Finance, particularly in relation to stablecoins, where the lack of transparency and real-time regulatory data heightens the risk of losses - as materialised in the collapse of the Terra Luna algorithmic stablecoin. The potential for significant economic and social harm is amplified by the cross-border nature of these markets, where regulatory arbitrage and jurisdictional gaps exacerbate risks.

Structural parallels with digital economy governance

The structural features generating regulatory blind spots in OTC derivatives and stablecoin markets – activities transcending regulatory perimeters, absence of mandatory disclosure, cross-border operation fragmenting supervision, and rapid scaling outpacing regulatory response capacity – replicate across the broader digital economy, particularly regarding platform services, artificial intelligence applications, and data intermediation. The degree of opacity mirrors the lack of transparency evident in derivatives markets before the Great Financial Crisis and in stablecoin markets more recently. Arguments that regulation would stifle innovation in digital sectors echo pre-crisis justifications for lighter touch regulation of the financial market. As the financial crisis demonstrated, however, under-regulation can result in socialised losses, where private gains are offset by broader societal harm.

These structural parallels suggest that regulatory responses addressing financial sector information asymmetries provide applicable templates, while requiring adaptation to digital characteristics. Mandatory reporting frameworks remain essential but must be expanded to create continuous data flows rather than periodic disclosures. Standardised taxonomies prove more challenging where activities and risks evolve rapidly, demanding continuous revision. Centralised repositories face technical challenges: data volumes, real-time processing requirements, and privacy considerations complicate aggregation mechanisms that proved effective for financial transaction reporting.

Amplifying features in digital contexts

Critical differences, however, constrain direct transposition of financial regulatory solutions. Several technological characteristics intensify situations of information asymmetry, as previously described, beyond the dimensions characterising financial markets.

Artificial intelligence applications generate information asymmetries qualitatively distinct from information that is not disclosed intentionally for strategic reasons. This is because machine learning processes produce outcomes not readily explicable through inspection of training data or model design. This "black box problem" (OECD, 2024, p. 17[26]) – where AI decision-making processes may remain partly opaque to developers and regulators alike – represents a different information challenge from that posed by complex financial instruments. Where derivatives exposures could theoretically be reconstructed through transaction records and accounting statements, algorithmic decision-making remains opaque even with full access to underlying algorithms and data. Financial regulation's gatekeeper mechanisms prove inadequate in this context: auditors and compliance functions lack technical capacity for substantive algorithmic assessment. The opacity of these systems requires methodologies that go beyond the enforcement of disclosure requirements but requires technical capacity for substantive algorithmic assessment that current regulatory institutions do not possess.

Platform operators accumulate comprehensive information regarding user behaviour, transaction patterns, and network interactions. Platforms' core business models depend upon proprietary data analytics inaccessible to external parties. This datafication operates in two directions: data aggregators and platforms – particularly the large technology firms – access and monetise extensive customer data relating to diverse areas of economic and social life, while financial service providers using such data can better assess individual clients' circumstances, including creditworthiness or propensity to purchase particular products. Simultaneously, the provision of digital financial services generates data that was previously proprietary to financial institutions. This data is particularly valuable as it is produced continuously and therefore provides longitudinal information on market participants. The various uses of this data remain largely invisible to regulators, who lack both the legal authority to compel disclosure and the technical capacity to analyse disclosed information meaningfully.

Supervision is more fragmented in respect of cross-border data flows than it is in relation to cross-border financial transactions. Financial institutions typically maintain legal entities or authorised branches in jurisdictions in which they operate, providing regulators with entities they can access for information requests and against whom they can take enforcement actions. Data processing, by contrast, occurs through distributed infrastructure spanning multiple jurisdictions without necessary correspondence to legal entity location. Cloud services enable instantaneous data transfer across borders, frustrating regulatory attempts to ascertain where data resides or which jurisdiction possesses authority to compel disclosure. This instantaneous and infrastructure-dependent character of cross-border data flows resists regulatory approaches designed for entity-based supervision, while the absence of physical presence eliminates the reciprocal enforcement mechanisms that characterised financial supervision.

Addressing these challenges requires multidisciplinary expertise – integrating technical, legal, economic, and sectoral knowledge – that exceeds regulators' already substantial capacity demands. Effective digital economy regulatory policy accordingly demands enhanced co-ordination mechanisms beyond those developed for financial services: real-time data-sharing infrastructure to replace periodic information exchange, standardised technical protocols for automated reporting verification, and legal gateways that allow authorities to grant cross-border supervisory access. The costs and political economy challenges prove substantial, explaining persistent co-ordination gaps despite widespread recognition of requirements.

Strategic intelligence and anticipatory capacity

Contemporary approaches to addressing information asymmetries increasingly emphasise the role of "strategic intelligence" – comprising horizon scanning, technology forecasting, and foresight exercises – to build governmental knowledge bases regarding the potential evolution of technologies and their impacts (OECD, 2025[3]). The absence of such forward-looking information capabilities leaves regulatory authorities insufficiently prepared when technological disruption occurs or when novel risk vectors emerge. Financial regulators' failure to anticipate the systemic implications of securitisation and OTC derivatives growth reflected not merely inadequate disclosure frameworks but the absence of institutional mechanisms for prospective risk assessment. Digital economy governance demands that regulatory authorities develop anticipatory capacities enabling identification of emerging information asymmetries before they grow large enough to matter in systemic terms – a requirement that current institutional setups are often not designed to fulfil.

Lessons for regulatory design and delivery

Establish comprehensive reporting arrangements and infrastructure prior to scaling

Information asymmetry, where private entities hold superior knowledge, leads to risk hidden in the system. The OTC derivatives and stablecoin cases demonstrate that the need for information is often underestimated until a crisis reveals a blind spot *ex post*. OTC derivatives achieved USD 600 trillion notional value before comprehensive reporting requirements emerged; stablecoins reached market capitalisations exceeding USD 100 billion in certain jurisdictions before some jurisdictions started conceiving reporting requirements. In both instances, there is significant cost of subsequent implementation.

Where market participants argue that disclosure obligations would impose excessive costs or compromise proprietary information, regulatory forbearance often prevails until a threat to the system becomes undeniable. However, information infrastructure established after activities achieve scale will always encounter resistance from entrenched interests. It will also face the challenge of implementing new practices across an active market of significant size. The regime under which regulation is shaped should include impact assessments that explicitly quantify potential risks and costs of non-intervention alongside compliance burdens.

Reflecting the strategic-intelligence capabilities discussed above, regulatory authorities should develop anticipatory capacities enabling identification of emerging information asymmetries before they achieve systemic significance (FSB, 2023[27]). Implementation constraints nevertheless prove substantial. Specifying reportable data elements for novel activities demands technical expertise regarding operational characteristics that may be imperfectly understood during early development phases. Over-inclusive requirements risk imposing excessive compliance costs on activities ultimately proving non-systemic, generating political resistance and potentially stifling beneficial innovation. Reporting obligations should therefore balance information-gathering against costs imposed on activities whose systemic significance remains uncertain.

Account for cross-border interconnections to avoid regulatory blind spots

In the financial market, global interconnectedness allows risks to spread across jurisdictions, often enabling market participants to exploit gaps in regulatory regimes. Firms avoid stricter oversight by operating in jurisdictions with lighter regulation. The same is true in the digital economy, where the global nature of services like cloud computing, social media and the provision of AI-related products and services, as well as cross-border data sharing, demand a cross-jurisdictional view to adequately capture relevant risks.

Effective cross-border gathering of information requires co-operation between regulators and supervisors. Market access conditionality – linking service provision to disclosure compliance – provides host jurisdictions leverage over foreign entities. Equivalence regimes condition cross-border access upon regulatory equivalence assessments. These create incentives for home jurisdictions to implement adequate data gathering frameworks. However, such approaches encounter limitations where services transcend territorial boundaries entirely. As digital platforms demonstrate, blocking access to remotely provided services proves technically challenging and politically contentious. On the other hand, it may be the case that the costs associated with separating operations by jurisdiction are relatively lower for digital goods compared to physical goods. Platforms can more easily generate multiple versions of their software, website or app, and they can offer different versions of their products to different customers, based on their increasingly sophisticated ability to use geo-identification technology to distinguish between customers located in different jurisdictions (Frankenreiter, 2021[28]), or even refrain from serving specific jurisdictions in response to regulatory interventions (OECD, 2024[29]).

Prioritise transparency and reporting to mitigate risks rooted in opacity

Information asymmetries arise from both strategic non-disclosure and inherent operational complexity. The financial market exhibited strategic opacity: bilateral relationships were invisible to third parties, and sophisticated participants systemically exploited information. The example of stablecoin issuance demonstrates similar dynamics: issuers resist transparency regarding the quality of the asset reserve and their capacity to redeem tokens at par, to avoid supervisory intervention or competitive disadvantages.

Algorithmic systems' opacity, however, rather reflects inherent complexity rather than strategic withholding. AI decision-making processes resist explanation to some extent even by developers, creating information deficits that disclosure requirements cannot remedy. Effective regulatory responses should therefore distinguish strategic from inherent opacity, deploying different mechanisms for each.

Whereas deliberate opacity can be mitigated through mandatory disclosure requirements and relevant enforcement, situations of inherent complexity demand the disclosure of system design, purpose and performance metrics. Ongoing monitoring processes could be needed to detect performance drift or other unintended effects. Implementation of both is highly resource-intensive, which raises the need for proportionality in line with risk.

3 Understanding new forms of systemic risk

Financial market regulation traditionally understood systemic risk as localised distress propagating through interconnected institutions and triggering widespread instability. Pre-crisis understanding emphasised sequential transmission through direct business relationships – a “domino” model of contagion wherein institutional failures cascade through counterparty exposures. This framework focussed on the risk that one participant’s failure to meet contractual obligations might cause other participants to default, creating chain reactions leading to broader final difficulties. However, this understanding proved inadequate in the context of the Great Financial Crisis and Digital Finance and is even less suitable when applied to the emerging digital economy.

The Great Financial Crisis revealed an inadequacy in respect of capturing contagion mechanisms operating through balance sheet interdependencies, market psychology, and confidence channels transcending direct counterparty relationships (Gorton and Metrick, 2012[13]). Instead, the crisis propagated through a collapse of the value of collateral that brought the repo market to a halt. (Pozsar et al., 2012[30]) established how shadow banking structures replicated traditional banking functions while operating outside prudential frameworks, creating systemic vulnerabilities invisible to supervisors focused on regulated institutions. The crisis exposed contagion through multiple channels simultaneously: linking institutions through common asset exposures and funding dependencies; psychological mechanisms wherein market participants’ collective behaviour amplified initial shocks; and opacity regarding risk concentrations that prevented effective supervisory intervention.

Systemic risk also propagates outside of purely financial transmission channels. Financial markets exhibit herding behaviour and panic dynamics analogous to traditional bank runs but manifesting through different mechanisms – asset fire sales, funding market freezes, and loss of confidence in counterparty creditworthiness (Avgouleas and Goodhart, 2015[31]). Behavioural finance research demonstrated how cognitive biases – loss aversion, overreaction to negative information, and reliance upon perceived actions of other market participants – generate self-fulfilling crises wherein market psychology deteriorates fundamentals independently of underlying asset quality (Shiller, 2003[32]). These psychological contagion channels operate with a speed exceeding that of traditional solvency-driven contagion, as markets react not merely to objective financial data but to perceptions regarding other participants’ likely responses. International co-ordination failures compounded these dynamics, as cross-border contagion occurred through globally integrated markets while regulatory responses were fragmented per jurisdiction (IMF, 2014[33]).

Contemporary regulatory frameworks therefore require concepts capable of addressing systemic risk arising from:

- operational dependencies upon digital infrastructure providers;
- psychological contagion amplified through information networks and algorithmic and AI-based trading;

- horizontal transmission across unrelated market segments through common exposures or correlated behaviour; and
- vertical propagation through value chains wherein disruption at single points cascades through dependent activities (Wilson, Sugimoto and Bains, 2022[34]).

In respect of digital financial services, this necessitates supervisory frameworks oriented toward network effects, concentration risks, and behavioural dynamics rather than exclusively institution-specific prudential supervision. Three case study clusters demonstrate how these expanded contagion mechanisms operate: first, market psychology driving the collapse of collateral value during the Great Financial Crisis; second, technological infrastructure failures propagating across dependent services, as illustrated by five recent information and communication technology incidents; and third, social media amplifying co-ordinated market movements, as demonstrated by the GameStop episode.

The three case studies reveal patterns in how systemic risk manifests through channels inadequately captured by traditional contagion models focusing on counterparty credit relationships. The acceleration, concentration, and psychological dimensions revealed across these diverse contexts indicate gaps in traditional supervisory approaches, suggesting that contemporary systemic risk demands regulatory frameworks addressing network effects, behavioural dynamics, and operational dependencies rather than focusing exclusively on institution-specific prudential metrics (FSB, 2023[27]).

Material regulatory rules are often direct products of a decades-long development of a specific understanding of “risk”. This applies equally to the institutional and procedural set-up of authorities. As a consequence, there is a significant possibility that risks or any form of unacceptable harm to societal values occur outside established categories of regulated risks. Equally, systemic propagation of risk may remain unidentified, as new contagion channels remain largely unexplored and are not yet established as objects of regulatory attention. The digital economy brings with it novel types of risk, such as discrimination by algorithms, platform-mediated misinformation, and cyber vulnerabilities affecting critical infrastructure. Regulatory policy frameworks should therefore incorporate mechanisms for continuous reassessment of risk taxonomies as technological and market conditions evolve (OECD, 2025[3]).

Case study 3: Uncertainty as a trigger

Market participants’ reactions are regularly influenced by cognitive biases including loss aversion and overreaction to negative news. During the Great Financial Crisis, behavioural finance mechanisms helped propagate risk from individual institutions to the broader financial system through channels distinct from traditional solvency transmission. This type of contagion spread faster than contagion involving actual solvency issues, as markets reacted not only to hard data but also to perceived actions and beliefs of other market participants. The feedback loops this created – wherein collective behaviour itself worsened underlying conditions – triggered negative asset-price spirals, causing values to deteriorate even faster than objective conditions warranted.

The Great Financial Crisis illustrates these psychological contagion mechanisms. The Crisis turned systemic when financial institutions began doubting valuations and credit ratings of mortgage-backed securities and collateralised debt obligations, realising the extent of subprime exposure. So-called “subprime” mortgages were repackaged (“securitised”) into structured products (“mortgage-backed securities”, or “MBS”) and sold to investors as seemingly risk-free assets with excellent credit ratings, relying heavily on the assumption that housing prices would continue to rise indefinitely. Often, investors repackaged MBS into new financial instruments, so-called collateralised debt obligations (“CDOs”) in a further securitisation, with no theoretical limit to how often this process could be repeated. When initially low interest rates began to rise, many subprime borrowers defaulted on their loans. As defaults increased, the value of MBSs and CDOs that had been widely distributed across the financial system plummeted, revealing the inherent risks that had been underestimated or ignored. Banks that held significant amounts of these products on their balance sheets saw a rapid deterioration in their capitalisation, leading to a loss of confidence among financial institutions.

Contagion occurred through multiple financial channels. However, beyond these, risk spread through market sentiment and psychology. As uncertainty grew regarding subprime-related asset values, financial institutions collectively withheld liquidity, fearing further exposure to toxic assets. This self-fulfilling behaviour created runs on liquidity, where aggregated individual actions amplified the overall crisis. Herding behaviour, driven by fear of losses in an opaque environment, exacerbated the liquidity crunch as institutions moved *en masse* to protect themselves, thereby intensifying systemic impact (Avgouleas and Goodhart, 2015[31]).

Case study 4: Concentration risk

Five recent high-profile information and communication technology failures have drawn attention to the combination of increasing outsourcing of critical functions to third parties, concentration of these functions in oligopolistic markets, and inherent lack of resilience amongst third-party service providers.

The 2016 SWIFT cyberattacks compromised local bank infrastructures by stealing valid member credentials to initiate fraudulent transfers through the trusted messaging system; in the Bangladesh central bank case, USD 81 million was stolen, with recovery only partially successful. The 2017 Amazon Web Services outage, triggered by human error during routine debugging, took critical servers offline, affecting numerous financial institutions – FinTech companies, payment processors, and banks – many of which lacked adequate redundancy owing to extensive reliance on a single provider, halting payment processing and online banking for several hours. The 2021 Facebook global outage severed communication channels for traditional and online-only financial institutions that relied on WhatsApp and Facebook Messenger for client support, authentication, and transaction confirmations, revealing operational dependency on non-financial platforms for essential customer-facing services. In 2024, a misconfiguration in Google Cloud's private cloud services deleted UniSuper's dedicated account, disrupting access for 600 000 Australian pension fund members for a week, though backups prevented data loss. Also in 2024, a faulty CrowdStrike update disrupted over 8.5 million Microsoft-based devices, causing system crashes and outages for banks, airlines, and healthcare facilities on a significant scale.

These incidents demonstrate that operational concentration creates systemic vulnerabilities (Wilson, Sugimoto and Bains, 2022[34]). Reliance on a small number of core providers – SWIFT for global payments, Amazon Web Services for cloud, CrowdStrike for cybersecurity – introduces single-point-of-failure problems whose effects cascade through interconnected markets instantaneously, both within a sector and across them. The transnational architecture of these services amplifies contagion: in the SWIFT breach, funds were siphoned across borders before detection, and the financial sector's reliance on real-time processing and continuous availability heightened its exposure. This concentration reflects highly concentrated market structures: institutions have few alternatives and cannot switch providers quickly given the scale and complexity of integration, so failure at a central provider spreads rapidly rather than remaining isolated (OECD, 2024, pp. 18-19[26]). The position is compounded because these firms, though technologically advanced, are not subject to the regulatory scrutiny applied to financial institutions, creating an imbalance in operational resilience across interconnected sectors.

Case study 5: Financial risk propagation through social media

The “GameStop short squeeze” in January 2021 was a financial event marked by a dramatic surge in the stock price of GameStop, a struggling United States video game retailer, due to organised efforts by retail investors to trigger a short squeeze. GameStop stock had been heavily shorted by hedge funds betting that its price would decline further. Retail traders, particularly those active on the Reddit forum r/WallStreetBets, noticed the unusually high shorting of the stock (reportedly over 140%) and initiated large-scale co-ordinated buying effort to drive up the stock price. The forum’s members shared information, memes, and emotionally charged rhetoric, rallying others to buy and hold GameStop stock in defiance of institutional investors. This dynamic extended to other platforms like Twitter and YouTube, where influential personalities further amplified the movement.

This resulted in GameStop's stock price rising from around USD 17 per share to a peak of USD 483 in a matter of days. As hedge funds rushed to cover their short positions by buying the stock to limit losses, upward pressure on the stock intensified. This dynamic caused significant financial losses for institutional short sellers. Retail traders, by contrast, perceived themselves as empowered, with many framing the event as collective action directed against institutional market participants.

Robinhood, a popular retail trading platform, played a critical role in the GameStop saga by providing infrastructure that enabled millions of retail investors to trade with no commission fees. However, on 28 January 2021, Robinhood controversially halted the ability to buy GameStop and other "meme stocks", citing liquidity constraints and the need to meet deposit requirements from its clearinghouses. This action angered many retail investors, as selling was still permitted, which contributed to sharp decline in GameStop's stock price.

Robinhood's actions prompted accusations of market manipulation, with allegations that the platform favoured institutional investors over retail traders. Several class action lawsuits were filed against Robinhood, and the event spurred calls for regulatory reviews regarding market stability, liquidity requirements, and transparency of trading platforms. Regulatory bodies, including the Securities and Exchange Commission, initiated investigations into whether these trading restrictions were justified and whether existing regulations governing broker-dealers and short selling were adequate. Concerns about hedge funds' influence, clearinghouses' role, and social media's impact on market volatility also surfaced.

The GameStop short squeeze was not merely a financial event; it was deeply intertwined with social dynamics, facilitated by online communities. Social media platforms (Reddit, Twitter) played a pivotal role in risk propagation, as the social cohesion they enabled created a feedback loop: as more traders joined the effort, the price continued to rise, inspiring even more individuals to participate. Online engagement allowed for rapid dissemination of information and co-ordination, challenging the traditional top-down market influence of financial institutions. The event highlighted how feedback between social media and retail trading can significantly move markets and create unexpected risks, potentially of even systemic dimensions.

Relevance for regulatory governance of the digital economy

First, all three case studies demonstrate contagion operating through mechanisms other than financial exposures – collective psychology, operational interdependence, and network effects rather than traditional solvency channels. Second, the cases reveal acceleration mechanisms that amplify initial shocks. Capital requirements force fire sales; institutions that depend on a common provider fail simultaneously rather than sequentially; and algorithmically reinforced co-ordination produces significant price movements within hours. Systemic risk in the digitalised economy therefore materialises at a speed exceeding historical precedents, (FSB, 2024, pp. 20-21[35]) constraining the time available for a regulatory response.

Third, structural concentration emerges as a vulnerability across all three cases, whereby ostensibly diversified or independent participants in fact share critical exposures, infrastructure, or execution channels – rendering those common elements suddenly systemically important. Fourth, in two of the case studies, psychological and behavioural mechanisms prove central to contagion dynamics. Herding behaviour – driven by loss aversion, uncertainty regarding others' likely actions, and social proof – generates self-fulfilling crises wherein collective responses deteriorate market conditions independently of the fundamental economics of the underlying assets.

Fifth, the cases demonstrate horizontal propagation across ostensibly unrelated market segments, through confidence channels and shared infrastructure dependencies rather than direct exposures. Systemic risk in the digital economy therefore propagates across sectoral boundaries rather than remaining confined within the perimeter of the financial system.

The channels through which systemic risk travels in the financial market experience are replicated, probably in a more pronounced manner, in the context of the digital economy. Several of its structural features amplify the relevant vulnerabilities, while introducing additional dimensions requiring regulatory attention.

The lack of resilience of digital infrastructure, even in the absence of high market concentration, intensifies the vulnerability at a single-point-of-failure (OECD, 2025[6]). Cloud services, cybersecurity systems, and application programming interfaces create dependencies wherein operational failures travel horizontally across sectors and vertically through value chains (FSB, 2024[35]). The economy's reliance upon a handful dominant cloud providers exceeds the dependence described in relation to the financial services sector (OECD, 2024, pp. 17,20[26]). Both create correlated exposures invisible to institution-specific supervision. The crucial difference lies in the scale and breadth of reach: whereas mortgage-backed securities involved hundreds of issuers and diverse structures, cloud infrastructure concentrates amongst a handful of dominant providers serving the majority of the market across all sectors of the economy simultaneously. Consequently, operational disruptions cascade not merely through the digital market but across commerce, healthcare, government services, communications infrastructure and finance concurrently.

Platformisation refers to the phenomenon of large, scalable networks of users that facilitate access to a broader range of financial and non-financial products and services. The phenomenon has two main components. The first is that platforms have a large, scalable digital network of users of services such as messaging services, social media, or personal device ecosystems. Within such digital networks the interaction between different users — be it consumers or businesses, or both — generates value for all participants. This value is amplified through network effects, where the utility of the platform increases as more users join and interact. The network effects inherent in platforms mean that each additional user increases the value of the network for all, creating a powerful incentive for new users to join, and disincentive for existing ones to leave. The original business model of platforms often relates to communication (Facebook), distribution (Amazon), device ecosystems (Apple), or data services (Google). The phenomenon also involves, as a second component, the offering of additional products and services different from the service on which the network's original success was built. In that respect, platforms are aggregators, combining own services with third-party services which they offer as agents or conduits, or merely through funnels. This market is dominated by BigTech firms. Despite the fact that only some directly provide financial services to customers in only few jurisdictions, they play a dominant role in the sector as providers of the relevant platforms and of the indispensable cloud services. BigTech firms control a significant part of the environment within which financial services are provided, and often participate in provision as a third-party provider, notably through cloud outsourcing arrangements.

Digital platforms' market structures concentrate transaction flows through limited intermediaries in ways exceeding traditional financial intermediation. Whereas financial markets involved numerous broker-dealers, clearinghouses, and settlement systems providing redundancy and competition, digital commerce increasingly channels through dominant platforms controlling search, recommendations, payments, and customer relationships. This concentration creates interdependency wherein policy decisions at platform level – algorithm changes, fee structures, data access restrictions – propagate across dependent businesses instantly. The parallel to clearinghouse dependency revealed during the GameStop episode proves instructive: Robinhood's trading restrictions demonstrated how platform operational decisions can move markets, but digital platforms exercise comparable influence across commerce, media, and services more broadly through algorithm and policy choices affecting millions of businesses simultaneously. The dependency of the economy on platforms therefore creates novel systemic vulnerability of a different order of magnitude as compared to the situation in the financial market.

The amplification of herding effects through algorithms represents a further dimension intensifying behavioural mechanisms observed in traditional financial contexts. The case study involving collateral values deteriorating as a consequence of market psychology demonstrated how loss aversion and herding behaviour generated self-fulfilling liquidity crises; social media platforms generally amplify these dynamics through content recommendation algorithms optimised for engagement rather than accuracy. Traditional bank runs required physical co-ordination and word-of-mouth propagation. Digital platforms now enable co-ordinated responses within minutes through algorithmically-curated information flows reaching millions simultaneously. The GameStop episode revealed this amplification operating in financial contexts, but comparable mechanisms pervade the digital economy more broadly – product reviews, service ratings, and reputational cascades all exhibit algorithmic acceleration of psychological contagion beyond historical precedents. Moreover, behavioural manipulation represents an additional systemic risk dimension unique to digital contexts: financial markets already exhibit herding behaviour and psychological contagion. Digital platforms go further, deploying algorithmic systems explicitly designed to influence user behaviour through personalised content, dynamic pricing, and limited choice (FSB, 2024, p. 26[35]).

Data concentration introduces systemic risk dimensions absent from traditional financial contexts. Financial institutions' mortgage-backed securities exposures created correlated risks through common asset holdings; digital platforms accumulate comprehensive data on individual behaviour, preferences, and transactions, creating a different form of data concentration. Data breaches, compromised data, or misuse of centralised datasets propagate across all services and institutions relying upon that data. This resembles but exceeds information asymmetry problems in financial regulation – whereas supervisors lacked information about shadow banking exposures, in digital contexts platform operators possess information asymmetries over both regulators and dependent businesses regarding market dynamics, consumer behaviour, and competitive conditions. The implications for systemic stability extend beyond market functioning, and ultimately to social cohesion.

Digital services achieve global market penetration at a speed that is substantially exceeding the diffusion of financial products (Wilson, Sugimoto and Bains, 2022[34]). App stores, social media networks, and influencer-driven adoption enable products to reach hundreds of millions of users across multiple jurisdictions within days of launch. This acceleration constrains regulatory capacity for risk assessment before novel services achieve systemic importance, as it shortens the time within which systemic dependencies emerge and entrench.

Regulation addressing systemic risk in the digital economy therefore requires new capabilities as compared to traditional supervision approaches. The digital economy presents concentration risks across data and infrastructure dependencies, algorithmic amplification of psychological contagion, intentional behavioural manipulation, and cross-sectoral propagation patterns exceeding those observed in financial markets. These characteristics demand regulatory innovation rather than transplantation of existing supervisory models.

Lessons for regulatory design and delivery

Identify critical points of failure across the digital economy to prevent systemic risks

Effective regulatory frameworks should identify critical points of failure across two dimensions: infrastructure providers (cloud computing, cybersecurity service providers, payment systems, and platform intermediation) and data aggregators whose datasets underpin services across multiple sectors. Infrastructure dependencies create operational dependencies wherein provider failures cascade across dependent institutions (Wilson, Sugimoto and Bains, 2022[34]); data concentration creates information asymmetries wherein platform operators possess superior knowledge of market dynamics, consumer behaviour, and competitive conditions relative to both regulators and dependent businesses. Designation criteria should address not merely market share but also substitutability, interconnectedness, and the extent to which concentrated resources create system-wide dependencies. However, imposing stringent requirements on dominant providers risks entrenching advantages by raising entry barriers for competitors, while light-touch regulation permits concentration to intensify – a dilemma particularly acute where network effects create winner-takes-most dynamics.

Recognise social contagion in digital markets and the role of algorithmic amplification

Regulatory policy could address both algorithmic amplification of psychological contagion and intentional behavioural manipulation through platform design. Amplification occurs where content recommendation systems optimised for engagement elevate emotions, thereby accelerating panic or herding beyond organic social dynamics.

Manipulation represents a distinct phenomenon wherein platforms deploy algorithmic systems explicitly designed to influence user behaviour through personalised content, dynamic pricing, and limited choice, creating systemic risk through co-ordinated alteration of individual decision-making. Transparency obligations could require disclosure of content ranking criteria, amplification mechanisms, and engagement optimisation objectives. Behavioural auditing could evaluate both whether algorithmic systems generally amplify harmful dynamics and whether design features intentionally exploit cognitive biases to alter user behaviour. Such an approach confronts fundamental tensions: prescribing specific algorithmic approaches risks entrenching incumbent systems and stifling beneficial innovation, while principles-based frameworks permit wide discretion potentially enabling both harmful amplification and manipulative design.

Prepare for horizontal and vertical propagation of risk across the digital economy

The case studies show that contagion operates horizontally across market segments and vertically through value chains. Regulation could address both propagation channels. Operational resilience requirements should impose stress testing accounting for simultaneous multi-sector disruptions rather than isolated institutional failures. Horizontal regulation proves more challenging to implement: sectoral regulation splits oversight across telecommunications, finance, commerce, and media while digital platforms traverse these boundaries, creating co-ordination problems and potential gaps.

4 Adjusting, repurposing and newly designing regulation

Financial regulation has traditionally operated upon assumptions of relative market stability, wherein rules, once established, would govern predictable institutional forms and business practices over extended periods. The approach of embedding regulation in prescriptive rules tied to specific market structures proves generally inadequate during periods of fundamental market transformation. The central theoretical problem concerns not merely regulatory content but regulatory form: static rule-based systems cannot accommodate the speed and unpredictability of innovation without generating either obsolescence (i.e., the rules fail to address novel practices) or ossification, i.e., outdated rules constrain beneficial development; see generally (Bennett Moses, 2007[36]).

Regulatory regimes operate as dynamic systems requiring continuous adaptation rather than periodic amendment of fixed rules. Effective regulation depends upon institutional mechanisms enabling regulators to monitor regulated activities, assess whether regulatory objectives are being achieved, and modify approaches accordingly. A distinction may be drawn between substantive requirements governing conduct and the institutional processes enabling rule adaptation. Where regulatory frameworks lack adaptive mechanisms, substantive rules inevitably lag behind market evolution, creating gaps that innovative actors exploit. Principle-based legislation is likely to be the most appropriate way of meeting policy objectives in complex or rapidly changing policy environments, though this remains contingent on the regulatory authority having the necessary sectoral expertise and capacity to implement its legislative responsibilities (OECD, 2012, p. 3[4]). The distinction proves critical: prescriptive rules-based systems optimise for predictability by specifying conduct requirements exhaustively, while principles-based approaches optimise for adaptability by articulating objectives and empowering regulators to develop implementation standards responsive to evolving circumstances.

The OECD's (2012) Recommendation on Regulatory Policy and Governance (OECD, 2012[4]) transforms these theoretical insights into institutional design principles. Principles 4 and 5 state that governments "integrate Regulatory Impact Assessment (RIA) into the early stages of the policy process" and "conduct systematic programme reviews of the stock of significant regulation", establishing mechanisms for continuous regulatory recalibration. Principle 7 requires that regulators possess "clearly defined mandates, roles and objectives" alongside authority to adapt implementation approaches as market conditions evolve. These provisions recognise that effective governance of dynamic markets requires not merely sound initial regulation but institutional infrastructure enabling anticipatory intervention and functional reinterpretation of existing instruments without comprehensive legislative reform. Agile regulatory governance requires mechanisms for continuous learning and adaptation as technologies and markets develop (OECD, 2024[37]; 2025[3]). The Financial Stability Board's (2011, 2014) post-crisis analysis of supervisory intensity demonstrates the practical necessity of such adaptive capacity: static capital requirements and entity-based supervision proved inadequate for addressing shadow banking and structured finance innovation precisely because regulators lacked authority to extend established principles to functionally equivalent activities operating outside traditional regulatory perimeters.

The case studies examined in this section reveal patterns wherein regulation designed for stable market environments proves inadequate when confronting significant transformation of market practice. Three phenomena account for this inadequacy. First, reactive regulatory approaches create delays between the moment when risk emerges and the relevant institutional response. Second, the mere formal compliance with rules enables the circumvention of the regulatory rationale in situations where regulation specifies requirements through prescriptive rules tied to legal classifications. Third, crisis response mechanisms improvised under time pressure are typically inadequate compared to sets of rules conceived *ex ante*. Consequently, regulatory effectiveness during periods of transformation depends upon procedural and institutional mechanisms enabling three adaptive capacities: forward-looking risk assessment that authorises intervention before harm materialises; functional reinterpretation of existing regulation to capture novel activities; and emergency protocols specified in advance so that competency is pre-allocated. Regulations lacking these second-order adaptive mechanisms prove incapable of governing dynamic markets regardless of first-order rule quality.

Case study 6: Lack of capacity to anticipate systemic risk

When Bear Stearns faced collapse in March 2008, the Federal Reserve provided emergency funding through its discount window, marking the first time since the Great Depression that the central bank had extended credit to a non-bank institution (Bernanke, 2010[38]). No US or European institution possessed the mandate or necessary capacities for *ex ante* systemic risk assessment. Federal Reserve Chairman Bernanke later testified that “for historical reasons, regulation and supervision were focused on the safety and soundness (or the practices) of individual financial institutions or markets. However, in the United States and most other advanced economies, no governmental entity had sufficient authority – now often called macroprudential authority – to take actions to limit systemic risks” (Bernanke, 2010, p. 3[38]). Lehman collapsed in September 2008 and five days later, Treasury Secretary Paulson proposed the Troubled Asset Relief Program, as no pre-established framework existed for addressing systemic liquidity shortages. For the EU, the de Larosière report stated that European financial supervision had operated “without any institution being specifically mandated to assess macro-prudential risks” (De Larosière et al., 2009, p. 37[39]; Goodhart, 2008[40]).

The institutional setup preceding the crisis reflected these gaps. The Financial Stability Forum, established in 1999 as a response to the Asian crisis, operated as a mechanism for co-ordination amongst G7 finance ministers and central bank governors. The FSF assessed vulnerabilities and promoted co-ordination in an ad hoc fashion, lacking competences for continuous monitoring or contingency planning for cross-border crisis management.

New systemic oversight institutions were created following the crisis: the European Systemic Risk Board in December 2010 and the US Financial Stability Oversight Council in July 2010 (see Regulation (EU) No. 1092/2010; (Dodd-Frank, 2010[41]) both receiving mandates for proactive risk assessment absent in the run-up to the crisis. The 2009 G20 London summit established the Financial Stability Board as successor to the Financial Stability Forum, with membership encompassing all G20 countries plus Spain and the European Commission, mandated to co-ordinate national authorities and international standard-setting bodies and to foster coherent implementation. Its competences include monitoring market developments, strategic reviews of policy work, and contingency planning for cross-border crisis management. Since 2011 it has conducted annual monitoring of non-bank financial intermediation covering 29 jurisdictions that represent approximately 80% of global GDP.

This change of institutional perspective and approach are intended to remedy the shortcomings in respect of the regulatory setup identified in the wake of the Great Financial Crisis. Pre-crisis regulatory setup lacked mechanisms for *ex ante* intervention. The evolution to macro-focussed financial regulation required a global financial crisis and two years of negotiation to upgrade from sporadic co-ordination to continuous monitoring, in itself illustrating again the institutional inertia that characterises reactive regulatory design.

Case study 7: Formalistic vs substantive focus of regulation

The financial technique of securitisation enables financial institutions to repackage illiquid assets, such as mortgages, into tradeable securities, thereby removing the relevant exposures from their balance sheets. The market for asset-backed securities (ABSs) grew from USD650 billion in 2004 to over USD 1.2 trillion by mid-2007 and collateralised debt obligations (CDOs) issuance expanded from USD 69 billion in 2000 to around USD 500 billion in 2006 (Gorton and Metrick, 2012[13]). The unusually high degree of attractiveness of these instruments can be explained by shortcomings of underlying accounting and capital rules. The capital framework of Basel II assigned risk weights as low as 7% for AAA-rated structured finance tranches compared to 100% for equivalent corporate exposures. Accounting standards under US GAAP FIN 46 and International Financial Reporting Standards permitted exemptions from consolidated accounting for qualifying special purpose entities, creating substantial capital arbitrage incentives. The effect of these rules was amplified by structured products' higher yields relative to comparably-rated traditional securities.

These rules enabled institutions to achieve off-balance-sheet treatment for structured finance and to circumvent regulatory limits on large exposures while maintaining minimal capital buffers against underlying risks. The regulatory framework hinged on formal elements in the relevant entities' accounting and standardised risk-weighting methodologies that assessed instruments based on structural characteristics rather than underlying economic substance, enabling market participants to achieve formal compliance while greater credit risk was contained within the relevant financial products than comparable traditional securities would contain. European banks had sponsored a substantial portion of US ABS issuance through off-balance-sheet special purpose vehicles. Major institutions including Citigroup, Merrill Lynch, and UBS participated in this market, while German Landesbanken accumulated significant structured finance exposures through conduit structures, with individual institutions in some cases showing exposure ratios exceeding 1 000% of capital.

When defaults on mortgages began to rise in 2006-07, structured finance markets experienced severe disruption as the performance of the mortgage portfolios contradicted rating agencies' risk assessment. Institutions suffered existential shocks: UBS had to write down more than USD 50 billion in subprime-related investments, Merrill Lynch had to be rescued by Bank of America, and most German Landesbanken collapsed, were rescued by peers or bailed out by the Government. Until the subprime crisis, they appeared well-capitalised under prevailing accounting standards (Financial Crisis Inquiry Commission, 2011[23]).

The unexpected and fast collapse of the market was a consequence of prescriptive capital rules tied to legal classifications that enabled circumvention of regulatory substance through merely formal compliance. The regulatory framework allowed market participants to assess financial products based on formal characteristics – credit ratings, legal structure, accounting treatment – rather than underlying economic substance. Countries such as New Zealand, Australia and the UK have, in the last five years, undertaken reviews of capital requirements to ensure that they remain proportionate and do not create competitive advantages for major banks while disadvantaging smaller banks and Non-Bank Financial Institutions (OECD, 2025[1]).

Case study 8: Improvised and pre-specified crisis response

Bear Stearns faced collapse in March 2008 when counterparties refused to extend overnight funding. The Federal Reserve orchestrated an emergency response over a single weekend, facilitating JPMorgan Chase's acquisition of Bear Stearns through a USD 29 billion credit facility secured against its least liquid assets, and invoking a Depression-era provision of the Federal Reserve Act unused since the 1930s (Financial Crisis Inquiry Commission, 2011, pp. 289-291[23]). No pre-established protocols existed: officials conducted continuous negotiations while simultaneously seeking legal justification for unprecedented central bank support to a non-bank institution, valued the distressed assets within 48 hours without standardised methodologies, and negotiated burden-sharing ad hoc (Bernanke, 2008[42]). The final terms resulted from an improvised compromise rather than from a pre-existing, generally applicable framework (Financial Crisis Inquiry Commission, 2011, pp. 290-291[23]).

Lehman Brothers encountered comparable liquidity pressures six months later, in September 2008, when short-term creditors withdrew funding and potential counterparties demanded additional collateral. Treasury and Federal Reserve officials again convened weekend negotiations, this time seeking either a private acquisition or consortium solution that would avoid public funding (Financial Crisis Inquiry Commission, 2011, pp. 324-339[23]). However, the improvised approach that had narrowly succeeded with Bear Stearns failed: no acquirer emerged willing to assume Lehman's liabilities without government guarantees, and authorities concluded that the Federal Reserve lacked adequate collateral to justify emergency loans. Absent pre-specified criteria governing when systemic risk warranted public intervention, and without a resolution framework applicable to systemically significant non-bank institutions, officials were unable to construct a viable rescue plan within the available timeframe. Lehman filed for bankruptcy on 15 September 2008 – an outcome that authorities had sought to avoid – triggering global market disruption (Financial Crisis Inquiry Commission, 2011, pp. 339–343[23]).

Silicon Valley Bank failed on 10 March 2023, triggering depositor withdrawals of USD 42 billion within 24 hours and marking the third-largest bank failure in United States history (FDIC, 2023[43]). With the overwhelming majority of deposits exceeding the standard USD 250 000 insurance limit, the risk of a run was acute (Federal Reserve, 2023[44]). On 12 March 2023, Treasury Secretary Yellen invoked the systemic risk exception under the Federal Deposit Insurance Act, enabling the FDIC to protect all depositors. That exception, introduced in 1991, follows a prescribed procedure requiring written recommendations from supermajorities of both the FDIC and Federal Reserve boards and a Treasury determination in consultation with the President; the relevant authorities determined that the failure posed systemic risk through contagion and broader economic spillover, particularly for small and medium-sized businesses (US Department of the Treasury 2023). The exception had been used only three times previously, during the 2008-09 crisis (FDIC, 2023[43]).

The contrast between this and the two previously described events is stark: SVB's resolution was organised on the basis of pre-specified exceptions centred on the prevention of systemic risk. It lacked the improvised character of the responses to the Bear Stearns and Lehman failures. The SVB case shows that effective crisis management is largely hinged on *ex ante* specification of mandates, processes and policy.

Relevance for regulatory governance of the digital economy

The structural features generating regulatory inadequacy in financial contexts – reactive approaches, formal compliance enabling the circumvention of substance ("box-ticking"), and improvised crisis response – manifest similarly in regulation for the digital economy, though often faster and with amplified consequences.

The reactive bias documented in the preceding case studies operates identically in digital contexts, and with higher speed. As Case study 6 shows, the evolution from the Financial Stability Forum to the Financial Stability Board required a global crisis and two years of negotiation; digital economy regulation confronts equivalent institutional design challenges while market structures entrench during the initial period. Regulation of artificial intelligence, platforms, or algorithmic content curation risks operating without forward-looking risk assessment analogous to post-crisis macro-prudential regulation. Such assessment depends on adequate information access – analysed in Section 2 – which digital economy regulators frequently lack.

The mechanism whereby the prescriptive rules of Basel II enabled box-ticking may similarly recur throughout digital economy regulation. This is typically the case where regulation defines the perimeter through entity categories or through referencing specific technologies. For instance, regulatory approaches organised around quantitative criteria for the designation of systemically relevant platforms, system classifications, or specific database technologies (e.g., "blockchain") replicate the vulnerability previously described: sophisticated actors may structure operations to achieve functional equivalence to regulated activities while exploiting classification boundaries. The specific manifestation differs but the underlying dynamic is comparable. Where regulatory perimeters depend upon categories of entities or technology classifications rather than functional outcomes, regulatory arbitrage is likely to arise.

The contrast between SVB's resolution through pre-specified mandates, processes and policy and the improvised responses to Bear Stearns and Lehman failures demonstrates that effective crisis management depends upon *ex ante* specification of authorities competent in an emergency situation, activation conditions, and co-ordination protocols. Digital economy regulation confronts equivalent challenges, notably in respect of the regulators' response to operational disruption (e.g., critical infrastructure failures, cybersecurity incidents) where rapid co-ordinated intervention across multiple authorities and jurisdictions is required. Where authorities improvise responses during incidents, they may encounter collective action problems, legal uncertainty, and co-ordination failures.

The overarching lesson concerns the inadequacy of static regulatory setup for governing dynamic systems. Financial regulation preceding the crisis embedded assumptions about stable market structures, predictable institutional forms, and evolutionary rather than revolutionary change. The digital economy presents similar challenges with amplified intensity: technological capabilities, business models, and risk transmission mechanisms evolve continuously while regulatory frameworks designed for earlier paradigms lag persistently. The case studies establish that effective governance requires not merely updating substantive rules to address novel practices but embedding institutional mechanisms enabling continuous adaptation. The OECD's Regulatory Policy Outlook 2025 emphasises the necessity of transitioning from "regulate-and-forget" approaches to "adapt-and-learn" processes that enable continuous improvement of regulatory systems (OECD, 2025[2]). This reflects the OECD's broader emphasis on anticipatory governance and adaptive regulatory mechanisms as essential elements for regulating rapidly evolving technologies (OECD, 2024[37]; 2025[3]). Path dependencies arising from delayed intervention – analysed in Section 5 – compound the costs of reactive regulation, as market structures and practices entrench during initial implementation periods.

Lessons for regulatory design and delivery

Anticipatory risk assessment

Regulatory frameworks for the digital economy lack established *ex ante* risk assessment capabilities analogous to the macro-prudential mandates established for financial regulation following the 2008 crisis. Effective regulation should authorise regulatory action based on forward-looking risk assessment, without requiring the prior occurrence of materialised adverse effects and incorporating lessons from past crisis events (OECD, 2012, pp. 19, 32[4]). A significant factor consists of constraints relating to the design of regulation: where the mandate specifies responsibilities depending on lists of regulated entities or activities, novel risks developing outside enumerated categories fall beyond regulatory reach until legislative amendment expands scope.

Effective frameworks require explicit statutory mandate for authorities to identify, assess, and address emerging risks transcending traditional regulatory categories (Black, 2005[22]). The OECD's Recommendation on Regulatory Policy and Governance (2012) proposes to put this approach into practice through requirements for regulatory stock reviews and integration of forward-looking assessment into policy processes. The OECD's 2021 Recommendation for Agile Regulatory Governance to Harness Innovation (OECD, 2021[45]) further advocates for the importance of anticipatory governance, incorporating systematic and co-ordinated horizon scanning and scenario analysis, anticipating and monitoring the regulatory implications of high-impact innovations, and fostering continuous learning and adaptation. Implementation requires dedicated analytical capacity for horizon-scanning and mandatory periodic reporting regarding identified emerging risks. Beyond that there is a need for transparent criteria governing the exercise of anticipatory powers to constrain arbitrary intervention. *Ex ante* intervention raises legitimate concerns regarding regulatory overreach. However, the experience of financial crises suggests that reactive approaches may generate significant social costs. The optimal balance requires combining *ex ante* risk assessment capability and the ability to adapt regulatory frameworks and oversight, and *ex post* accountability.

Functional reinterpretation of regulatory instruments

Where regulation in respect of the digital economy depends upon rigid classifications or categories rather than functional outcomes, the risk for regulatory arbitrage rises. Regulation could incorporate explicit authority for regulators to extend existing instruments to activities that are functionally equivalent to regulated activities which operate outside formal regulatory classifications. Functionally equivalent activities, i.e., activities producing similar economic effects, should receive consistent regulatory treatment regardless of legal form or technological implementation. This understanding enables regulatory gaps to be filled without comprehensive overhaul of the rulebook (FSB, 2023, p. 2[27]). A more principles-based framework allows for an increased degree of adaptability by articulating objectives and empowering regulators to develop responsive implementation standards.

Procedural safeguards should accompany this authority: specific criteria should determine the circumstances under which functional equivalence justifies an extension of regulation to new market practices. To determine equivalence a sophisticated assessment of economic substance is necessary, detached from relevant categories or classifications (FSB, 2023[27]) Critics argue that principles-based frameworks create excessive regulatory discretion, but experience demonstrates that prescriptive approaches prove generally inadequate for dynamic markets (Black, 2008[46]): the choice concerns not whether gaps emerge but whether regulation enables gaps to be addressed through administrative adaptation, or requires comprehensive legislative amendment.

Pre-specified emergency frameworks

Regulatory policy for the digital economy has to address the possibility of operational disruption with systemic effects and be able to adequately respond to it. This may require rapid co-ordinated intervention across multiple authorities and jurisdictions. However, most rulebooks lack pre-defined protocols: authorities improvise responses during incidents, encountering collective action problems, legal uncertainty, and co-ordination costs. The SVB resolution through pre-specified emergency procedures, contrasted with the improvised Bear Stearns and Lehman responses, demonstrates that co-ordination costs during improvised responses generally exceed constraints arising from pre-specified protocols. Pre-specified protocols are part of anticipatory regulatory design, as they are developed and tested before crises materialise (OECD, 2024[37]).

Regulation should specify emergency intervention authorities, activation conditions, and decision-making procedures *ex ante*. This requires:

- legislative or administrative specification of circumstances warranting public sector emergency intervention, with quantitative or qualitative thresholds (FSB, 2014, revised April 2024[47]), Key Attribute 3.1; I-Annex 4, para 4.1(i));
- allocation of decision-making authority among relevant agencies, with clear hierarchy preventing co-ordination deadlock;
- mandatory information-sharing obligations with specified protocols enabling rapid data exchange; and
- sunset provisions ensuring return to regular supervisory frameworks once immediate threats subside.

Rigid intervention thresholds may prove over-inclusive or under-inclusive in particular circumstances. However, the benefits of institutional clarity and reduced co-ordination costs under crisis conditions probably outweigh the relevant cost.

Periodic perimeter reassessment

Rulebooks should incorporate processes for periodic reassessment of whether the regulatory perimeter adequately captures activities performing regulated functions (Sunstein, 2011[12]). A proactive review allows for continuous adaptation that is essential for an effective governance of the digital economy

(OECD, 2025[2]). The OECD's analysis emphasises that built-in review and adaptation mechanisms are essential for ensuring regulations remain effective as technologies develop (Sarliève et al., 2025[48]). While the application of rules on the basis of functional interpretation enables the filling of regulatory gaps, periodic reassessment establishes institutional processes for *ex ante* perimeter adjustment before regulatory gaps allow for material harm to occur.

The implementation of this lesson requires mandatory periodic review of the regulatory scope – the OECD's Recommendation on Regulatory Policy and Governance (OECD, 2012[4]) operationalises this through requirements for "systematic programme reviews of the stock of significant regulation". Going beyond this, regulatory reviews could explicitly address whether activities that are functionally equivalent to regulated activities operate outside regulatory reach and assess whether gaps create material risks to regulatory objectives.

5 Innovation and path dependency

Every regulatory response to innovation entails choices that naturally constrain future policy options. To support innovation and competitiveness or react to crises, in particular, regulators may, actively or through inaction, create conditions that encourage the adoption of specific market practices, including intended as well as unintended ones. Their decisions naturally create effects, some of which may be difficult to reverse. In this context, path dependency explains how initial regulatory choices set parameters for future options, pre-shaping developments that limit flexibility to respond to new conditions or information as they arise (Black, 2005[22]).

Thus, regulators and markets can become locked by these conditions into a development that is difficult to influence and correct, even as relevant risks become apparent. The entrenchment of decisions constrains future policy making, due to not only the costs but also the institutional and structural complexities of reversal. The ongoing effects from early regulatory decisions then cause a narrowing of the scope available for policy making, leading to increasingly reduced flexibility to adapt to new or unforeseen challenges.

Two case studies examined below reveal a common causal mechanism through which regulatory choices – whether active intervention or deliberate forbearance – generate path dependencies constraining future policy options. In each case the path dependency operates through three stages, notably the establishment of an initial regulatory position, followed by market entrenchment, and subsequent resistance of incumbents to regulatory changes. Hence, regulatory forbearance may generate path dependencies constraining future policy, with reversal costs escalating significantly as markets entrench around initial regulatory frameworks.

Case study 9: The difficult rolling back of the wheel

In the run up to the Great Financial Crisis, the market practice of securitisation and the use of repurchase agreements (repos) became central to financial market operations. Securitisation, particularly of "subprime" mortgage portfolios, allowed financial institutions to convert otherwise illiquid assets with an opaque risk profile into tradable securities of notionally low risk. Through this process, banks could offload relevant risks to investors and remove them from their balance sheets. However, substantial residual risks remained with the banks themselves, and the risk of reputational damages if a securitised asset failed remained largely unaccounted for (Goodhart, 2008[40]).

Credit ratings were central to this process. Mortgage-backed securities (MBSs) and collateralised debt obligations (CDOs) obtained top ratings from Standard & Poor's, Moody's, and Fitch and were, as a consequence, subject to relatively low capital requirements under Basel II, which made them attractive for banks to hold and positioned them ideally as collateral in repo transactions. In the years before the crisis, repos allowed institutions to increase leverage dramatically: virtually all major institutions borrowed heavily against MBS and CDO collateral in a repo market that expanded to some USD 10 trillion by 2007. Because repo financing depended on the credit rating of the collateral, the system was rendered highly liquid but fragile – every decline in asset values automatically reduced collateral value, impairing the availability of repo funding and firms' balance sheets.

Retrospective analysis revealed that rating agencies had frequently and significantly understated the risks inherent in structured products like MBSs and CDOs (Turner, 2009, pp. 22-23, 77[14]). The effects on banks and the financial system at the start of the Great Financial Crisis were devastating as credit ratings were firmly embedded not only in market practices but also in the regulatory framework itself. Notably, Basel II allowed banks to rely on credit ratings to determine the capital they needed to hold against different types of assets. This framework incentivised issuers of structured financial products to maximise the products' credit ratings, irrespective of whether these ratings accurately reflected the underlying risk. Rating agencies failed to provide realistic ratings, thereby implanting hidden risk throughout the banking system.

These flaws were visible and could have been addressed before the Great Financial Crisis. However, even in the crisis's wake, the system-wide reliance on credit ratings continued. Financial institutions had become dependent on credit ratings for their profitability and liquidity. The financial system relied on this high level of liquidity, meaning that altering the rating-based framework risked causing systemic instability. In the 15 years since the crisis, reforms have been limited in scope, with changes proceeding only gradually and incrementally. Reversal resistance manifested through multiple reinforcing mechanisms: economically, no immediate alternatives existed for credit ratings' risk assessment function; institutionally, financial institutions had developed sunk investments in compliance infrastructures and operational processes, rendering regulatory reversal prohibitively costly; politically, rating agencies resisted structural changes to business models.

Case study 10: Diverging regimes for similar risks

As of 2009, crypto-assets attracted an increasing number of market participants, starting with a small group of tech enthusiasts and growing to a market capitalisation of approximately USD 1 trillion today (though it fluctuates heavily based on market conditions). No dedicated regulation was introduced in the initial phase, as crypto-assets did not constitute a regulated financial service or product or any other regulated activity. However, as the risk of criminals and corrupt regimes using crypto-assets to hide and transfer value started to emerge, existing rules on anti-money laundering and counter-terrorist financing were clarified, under the auspices of the Financial Action Task Force (FATF), to apply to crypto-assets (Simmons, 2021[49]), and many jurisdictions adjusted their tax regimes as well to mitigate avoidance.

This changed when, in 2015, New York's Department of Financial Services introduced the BitLicense. This regulatory framework required businesses dealing with crypto-assets to obtain a licence and adhere to specific disclosure and reporting standards. For the first time, regulation specific to crypto-assets was introduced, based on the limited rationale of consumer protection, primarily using the regulatory tools of market access regulation and disclosure.

As the crypto-asset market grew, some argued for its more comprehensive regulation. Broad consensus emerged that services or products functionally corresponding to regulated financial services or products were covered by existing regulation. In particular, shares and bonds could not evade relevant securities laws simply because they were recorded and evidenced via a blockchain database. Consequently, existing securities laws generally apply to these types of assets.

In addition, the FSB published principles regarding global stablecoins, products that mimic the function and stability of money and are widely adopted. There is no such unanimity when it comes to other forms of crypto-assets, including those that only remotely resemble financial products or services, including Bitcoin.

In 2023, the EU comprehensively regulated those crypto-assets which were not already regulated as shares or bonds, introducing the Markets in Crypto Asset Regulation (MiCAR). Under MiCAR, stablecoins are subjected to stringent rules on reserves, governance, and systemic impacts. Crypto-assets not resembling any financial product or service, such as Bitcoin, are subjected to a light market access regime and rules focused on consumer protection. Crypto-asset service providers, such as wallets and exchanges, are regulated similarly to traditional securities service providers, such as brokers and stock exchanges.

The initial absence of crypto-asset regulation (other than that mitigating illicit activity) permitted a staggering growth of that market outside the regulatory perimeter. In many cases this was a deliberate policy choice to promote financial innovation and market competitiveness. The market structures thus created prove relatively resistant to subsequent regulatory intervention.

The EU's MiCAR exemplifies the resulting constraint: by 2023, crypto-markets had achieved sufficient scale that comprehensive regulation required accommodating rather than reshaping established market structures. MiCAR's technology-specific approach locked regulation into particular technical setups rather than functional outcomes. As a consequence, a market emerges with risks comparable to that of the financial market which, however, is subjected to different regulatory standards.

In the US, in 2025 the GENIUS act has been adopted on the regulation of stablecoins, which largely reflects best practice market standards without introducing significant limitations on the industry, thereby enabling the industry to interact with the regulated financial system. Legislative proposals for the regulation of crypto-asset service providers are still in the legislative process at the time of writing.

Relevance for regulatory governance of the digital economy

Path dependency operates not merely through technical lock-in but often through deliberate regulatory choices that shape market structures or enable them to entrench. The structural patterns generating path dependencies in financial regulation manifest similarly in digital economy contexts.

The concentration of digital infrastructure creates structural dependencies comparable to the dominance of credit rating agencies: financial institutions, healthcare systems, government administrations, and commercial sectors simultaneously depend to a significant extent on these. Any subsequently conceived regulatory framework has to factor in reversal costs if it is designed to change things, such as migration complexity, service interruption risks, and interoperability challenges. Where the reliance on credit rating developed over decades, digital platforms can achieve comparable dominance within a few years. Relevant changes in market practice occur within significantly shortened timeframes, as opposed to a market where regulators were historically able to formulate responses before the relevant problematic practices solidified.

Technology-specific regulation can generate path dependencies by embedding particular technological assumptions within the legal framework. For instance, MiCAR's focus on distributed ledger technology may create regulatory fragmentation, as functionally equivalent activities are treated differently depending on their technical configuration. Such approaches can create opportunities for regulatory arbitrage through strategic technical design choices. These vulnerabilities are especially pronounced in rapidly evolving markets, with the risk that rules become obsolete as technology evolves.

Supporting innovation and regulating to protect from risks are often presented as binary choices, whereas the reality is more nuanced. Markets also require stable and predictable conditions to attract investment. Pre-crisis financial regulation relied on the financial market's self-regulatory forces to ensure sufficiently prudent practices in innovative markets, such as the securitisation market. Recent debates on the regulation of digital markets regarding platform regulation, data governance, and AI oversight demonstrate the politicisation of this topic, bringing in arguments over competition and debates over values such as free speech. In both examples, market structures have developed fast and solidified with relative freedom (FSB, 2023[27]), thereby increasing the cost of subsequent regulatory intervention.

Particularly in a market environment characterised by disruptive practices, the setting of intervention thresholds as well as the determination of intensity of regulation has significant future effects on incumbents and new entrants, potentially initiating a path-dependent development of the market once the initial regulatory reaction has nudged the market in one or the other direction (Pozsar et al., 2012[30]). Strategic foresight methodologies, which develop alternative future scenarios and their policy implications, provide instruments for assessing the potential long-term effects of regulatory choices (OECD, 2025[3]). As part of anticipatory governance, such approaches inform regulatory choices before technological developments constrain available policy options, expanding policymakers' capacity to anticipate how market developments may interact with those choices well into the future.

Lessons for regulatory design and delivery

Foresee regulatory flexibility

Regulation for digital markets should be designed to accommodate technological change. Rules focused on what market participants do and the effects thus created should be flexible enough to adapt, allowing regulators to adjust requirements as markets develop, without requiring new legislation for each technological shift. Such "outcomes-based regulation" focuses on achieving or avoiding specified results while affording regulated entities flexibility in determining the means of compliance (OECD, 2012[4]; 2024[37]; 2025[3]).

Long-term consequences

Policy makers should be mindful of the timing issues around when to regulate, taking into account the potential creation of path dependencies. Concentration around key infrastructure providers or dominant platforms, once established, is difficult to reverse.

Regulators should therefore be empowered to act early, before harm has materialised, when indicators suggest that a market is growing rapidly or becoming concentrated. Where uncertainty remains about how a market will develop, regulation can be introduced in stages, with requirements adjusted as the situation becomes clearer.

Institutional design matters. The functions of promoting industry and supervising it should be separated, so that promotional objectives do not override supervisory judgement (OECD, 2012[4]), Principle 7). Regular review of unregulated market segments should be mandatory, supported by mechanisms that support horizon scanning and strategic foresight methods to implement anticipatory regulatory approaches, enabling identification of regulatory gaps before harms materialise (OECD 2025a, 20, 61).

Reversal of regulatory choices

Once markets develop around a particular regulatory approach, change becomes difficult. Firms invest in compliance systems built for existing rules, alternatives to established practices may not exist, and incumbent firms resist reforms that threaten their position. These factors compound over time, making any correction of regulation progressively harder.

Where opportunities for early intervention have been missed, regulators should at a minimum ensure that new market entrants and incumbents are subject to equal standards (OECD, 2021[50]). This prevents the competitive advantage that would otherwise accrue to incumbents whose business models developed under permissive conditions (OECD, 2021[50]).

6 Regulatory effectiveness

Effective regulation depends upon institutional foundations encompassing clear mandates and objectives, adequate resources and capacities, appropriate methodologies for risk-based decision-making, transparent enforcement frameworks, and co-ordination mechanisms where multiple authorities share responsibilities. Each of these dimensions proves necessary but not enough on its own; their interaction determines whether regulatory authorities can perform their role and fulfil their mandate effectively. The financial crisis of 2007-09 exposed failures across all five dimensions (Turner, 2009[14]; FSB, 2014[51]). Subsequent developments, in particular the emergence of large technology platforms as service providers in the financial market, have intensified these challenges.

Case study 11: Institutional design and clarity of the mandate

The effective discharge of regulatory responsibilities depends upon clear statutory specification of objectives and powers. Ambiguity in respect of the mandate generates uncertainty regarding the scope and limits of regulatory authority, potentially leading to regulatory gaps or, conversely, excessive intervention beyond intended boundaries (Ferran, 2012[52]). At the same time, institutional arrangements influence effectiveness. The allocation of responsibilities across multiple authorities creates co-ordination challenges, particularly where regulated activities span traditional sectoral boundaries or where systemic risks require comprehensive oversight transcending individual institutional mandates.

The United Kingdom's pre-crisis tripartite regulatory setup exemplifies the risks inherent in fragmented institutional structures and ambiguous mandates. Established in 1997, this arrangement divided responsibilities amongst the Financial Services Authority (prudential and conduct supervision), the Bank of England (monetary policy and financial stability oversight), and the Treasury (regulatory framework design). The framework rested upon a Memorandum of Understanding specifying co-ordination mechanisms yet lacked binding enforcement provisions or clear demarcation of crisis management responsibilities.

The collapse of Northern Rock in September 2007 exposed deficiencies in this setup. The institution's failure – which caused the first bank run in the United Kingdom in over a century – revealed co-ordination failures arising from ambiguous mandates and voluntary co-operation protocols (Goodhart, 2008[40]). Specifically, the FSA's mandate for micro-prudential supervision was interpreted narrowly as firm-specific oversight, with systemic stability considerations falling outside its operational focus. The Bank of England possessed a general financial stability mandate but lacked direct supervisory powers over individual institutions and therefore possessed insufficient real-time information regarding Northern Rock's deteriorating position. When the institution approached the Bank of England seeking emergency liquidity assistance, the Bank of England faced uncertainty regarding both the scale of Northern Rock's funding gap and the legal boundaries of its lender-of-last-resort function absent imminent systemic threat. The Treasury, while responsible for the legislative framework and deposit insurance, had not updated the depositor protection scheme to address wholesale-funded institutions vulnerable to confidence shocks, and statutory provisions granted no clear authority to intervene pre-emptively in troubled institutions.

These ambiguities regarding the mandate translated into operational failures. The FSA had identified relevant risks in Northern Rock's business model as early as 2006 but concluded these were acceptable given the institution's regulatory capital levels, as the FSA's supervisory guidance emphasised capital adequacy but neglected the resilience of a bank's funding structure. The Bank of England, monitoring wholesale funding markets, observed general tightening during summer 2007 but lacked institution-specific data to assess Northern Rock's vulnerabilities and possessed no clear mandate to demand such information from the FSA. The subsequent "weekend crisis" exposed the absence of pre-defined co-ordination protocols: no authority held clear responsibility for assessing whether emergency liquidity assistance should be provided, under what terms, with what disclosure requirements, or with what implications for depositor protection. Emergency legislation authorising unlimited depositor guarantees came only after queues of depositors had formed outside Northern Rock branches.

The Turner Review's subsequent analysis concluded that mandate fragmentation had created accountability gaps, with each authority assuming others were monitoring systemic risks while none held clear responsibility for comprehensive oversight (Turner, 2009[14]; Ferran, 2012[52]). The Memorandum of Understanding between FSA, BoE and Treasury had specified information-sharing protocols but could not compel co-operation under stress, when each institution prioritised its own statutory objectives and reputational concerns. The FSA's narrow interpretation of its micro-prudential mandate reflected rational organisational behaviour given resource constraints and absence of explicit systemic stability responsibilities. The Bank's limited intervention reflected both legal uncertainty regarding its powers and information deficits arising from its exclusion from direct supervision. The Treasury's reactive approach reflected an approach rooted in a statutory framework designed for traditional deposit-taking institutions rather than wholesale-funded mortgage lenders.

The United Kingdom subsequently abolished the tripartite structure in 2013, transferring prudential supervision to the Prudential Regulation Authority (a Bank of England subsidiary) with an explicit financial stability mandate, and conduct supervision to the Financial Conduct Authority, thereby implementing an objectives-based "twin peaks" model designed to clarify each organisation's responsibilities and reduce co-ordination costs.

The described failure demonstrates that voluntary co-ordination protocols, however detailed, prove insufficient under stress. Without binding mechanisms to compel information sharing and joint decision-making, institutional self-interest and divergent priorities may fragment crisis responses. Moreover, ambiguity of the relevant mandate permits regulatory gaps to emerge not through explicit legislative omission but through each authority's rational interpretation of its responsibilities as excluding matters assumed to fall within others' remit. Effective institutional design therefore requires not merely allocating responsibilities across authorities but specifying those responsibilities with sufficient precision to eliminate gaps, establishing binding co-ordination mechanisms with enforcement provisions, and creating clear accountability mechanisms for comprehensive oversight where systemic risk concerns the mandates of more than one authority. The legislative framework should explicitly address such boundary cases rather than relying upon informal ways to resolve ambiguities arising during crises.

Case study 12: Resource constraints and regulatory capacity

Much like the public sector in general, supervisory authorities only dispose of constrained capacity, arising from limited budgets and difficulties recruiting and retaining specialist expertise, especially given the competition with private sector salaries for specialists in finance and technology.

The UK FSA's situation with regard to the Northern Rock case illustrates the resulting gap between mandate and capacity. Established as an integrated regulator with comprehensive responsibilities spanning banking, insurance, and securities regulation, the FSA confronted an exceptionally broad remit encompassing both prudential and conduct objectives across diverse sectors. Throughout the 2000s, the FSA adopted a supervisory philosophy emphasising reliance upon firms' internal risk management capabilities rather than intensive regulatory oversight, termed "risk-based supervision". This approach reflected a policy preference for market discipline. However, it was also grounded in practical constraints: the FSA's resources proved inadequate for comprehensive supervision of all institutions within its remit.

Risk-based supervision, while conceptually sound as a resource allocation methodology became a vehicle for spreading limited capacity across an enlarged regulatory scope, thereby diluting overall supervision intensity. The FSA allocated supervisory resources according to assessed probability and impact of firm failure, concentrating attention upon systemically significant institutions. However, resource limitations prevented even high-impact firms from receiving sufficiently intensive oversight. Northern Rock, classified as a high-impact institution, was supervised by a small team lacking the capacity for continuous monitoring of the bank's rapidly evolving business model and funding vulnerabilities.

Resource constraints impeded effective financial supervision. The Financial Stability Board found that prior to the crisis, supervisory frameworks had become "very light touch", with "insufficient attention... paid to supervisory skills" and "specialist skills... in short supply" (FSB, 2014, p. 2[51]). The FSA's 2008 internal audit of its Northern Rock supervision identified inadequate supervisory resources as a central failure, concluding that finite resources result in many firms receiving very little supervisory attention (FSA, 2008[53]; Turner, 2009[14]). Capacity constraints manifested across multiple dimensions. First, staffing levels proved insufficient relative to the number and complexity of supervised institutions. The FSA supervised over 27 000 authorised firms with approximately 2 700 staff, a ratio precluding intensive engagement with most entities. Second, expertise gaps limited the Authority's ability to assess sophisticated structured finance products and wholesale funding strategies. Third, high staff turnover, driven by different levels of compensation compared to the private sector, resulted in loss of accumulated institutional knowledge and supervisory experience.

Resource constraints contribute to supervisory failure, especially in combination with other inefficiencies within the supervisory setup. Supervisory frameworks designed to optimise the allocation of resources can become mechanisms for rationalising inadequate resourcing, thus serving to justify insufficient attention to certain groups of, or even all, supervised entities. Where capacity proves inadequate for the required levels of supervision, risk assessment frameworks cannot compensate; they merely systematise under-supervision (FSB, 2014[51]).

Case study 13: Supervisory methodologies

Supervisory methodologies seek to allocate scarce supervisory resources according to the assessed probability and potential impact of regulated firms' non-compliance or failure. They translate regulators' statutory objectives into operational risk categories, develop metrics for measuring risks within each category, and link resource allocation to overall riskiness. When effectively implemented, and provided that sufficient resources are available, the right methodology enables regulators to prioritise attention upon areas posing greatest threats to their objectives, deploy specialist expertise efficiently, and provide transparent criteria for supervisory interventions.

In the years prior to the financial crisis, the UK FSA's "Advanced Risk-Responsive Operating Framework" (ARROW), introduced in 2003, was meant to achieve these goals through the methodology of "risk-based supervision". ARROW required supervisors to assess both inherent risks in firms' business models and the adequacy of control systems addressing those risks. Assessment encompassed ten high-level risk categories spanning business risks (customers, products and markets; business processes; prudential risks) and control risks (market conduct controls; financial and operating controls; prudential risk controls). Supervisors assigned probability and impact scores, combining these into overall risk classifications determining supervisory intensity.

However, as discussed in the previous section, the FSA lacked both the technical skills and the time required for penetrating analysis of highly complex business models and risk exposures. Second, the framework created "dark holes" – risk areas either left unscored or assigned default medium ratings when supervisors lacked information to make confident assessments. Northern Rock's risk profile contained multiple such gaps, particularly regarding the important assessment of the sustainability of its wholesale funding model under stress conditions. Third, ARROW's reliance upon firms' self-assessment and management information meant that supervisors' risk evaluations depended heavily upon data provided by the supervised entities themselves. This created information asymmetries favouring firms, particularly where management had incentives to understate risks or overstate control effectiveness (see Section 2 on information asymmetries). Fourth, risk factors changing with the market environment – i.e., developments affecting multiple institutions – proved difficult to integrate into firm-specific assessments. Even though the FSA maintained specialist teams for horizon-scanning and sectoral analysis, translating risks identified at macro-level into adjustments of individual firms' risk scores encountered significant obstacles.

Following Northern Rock's collapse, the FSA introduced "intensive supervision" for high-impact, high-risk firms, acknowledging that ARROW's standard approach provided insufficient oversight for systemically significant institutions. Generally, risk-based frameworks require continuous updating to reflect evolving market conditions, regular training to ensure consistency in applying relevant criteria, and robust processes ensuring quality. Absent these supporting mechanisms, risk-based and similar methodologies can create false confidence in supervision while its actual effectiveness remains inadequate (FSA, 2008[53]).

Case study 14: Intervention thresholds and enforcement discretion

Regulatory effectiveness also depends upon the timing of intervention. Clear specification of intervention thresholds – the circumstances triggering action – is essential for effective supervision. Ambiguity regarding when regulators may or must intervene creates two distinct pathologies: either premature intervention that courts subsequently invalidate for exceeding statutory authority, or delayed intervention reflecting regulators' uncertainty regarding their legal powers or fear of judicial reversal.

Germany's banking supervision framework, while granting the supervisor BaFin extensive powers, historically provided less clarity regarding the specific circumstances warranting intervention. IKB Deutsche Industriebank's distress during 2007 demonstrated the consequences. IKB, a mid-sized German bank specialising in lending to medium-sized enterprises, had accumulated substantial exposure to US subprime mortgage-backed securities through a conduit structure. When these assets deteriorated in value during 2007, IKB faced potential insolvency. BaFin's intervention came only after the crisis materialised, rather than through early action to constrain IKB's risk accumulation. Post-crisis analysis identified hesitation to intervene absent clear evidence of imminent failure as a contributing factor. BaFin possessed information regarding IKB's exposures but lacked institutional confidence to impose binding constraints on the bank's strategy before its regulatory capital was impaired (Swaminathan et al., 2024, p. 157[54]).

Similarly, Germany's Hypo Real Estate's 2008 near-collapse revealed co-ordination gaps between BaFin and the Bundesbank, the German central bank (which conducts ongoing supervision on BaFin's behalf pursuant to the KWG's allocation of responsibilities). The division of responsibilities between these institutions created potential for delays and diluted accountability, as occurred when HRE's liquidity crisis emerged in September 2008 following Lehman Brothers' failure. The subsequent German financial stabilisation legislation and reforms to BaFin's governance sought to address these deficiencies by clarifying intervention powers and establishing more explicit co-ordination protocols.

More recently, the Wirecard scandal (2020) exposed persistent gaps in BaFin's supervisory effectiveness and intervention capacity. Wirecard, a payments processor, collapsed following revelations of massive accounting fraud. BaFin's mandate focused primarily upon Wirecard's banking subsidiary, while the (non-bank) parent was not directly supervised by it, creating supervisory blind spots. Moreover, when concerns emerged regarding Wirecard's accounting, BaFin initially took enforcement action against journalists and short-sellers who raised questions rather than intensifying scrutiny of the company itself – a response reflecting institutional capture or misaligned priorities. The episode demonstrated that regulatory effectiveness depends not merely upon formal powers but upon willingness to deploy them against influential institutions, and upon the existence of an institutional culture prioritising systemic stability, market integrity and investor over sectoral reputation.

The specification of intervention thresholds requires balancing competing considerations. Overly prescriptive thresholds risk mechanical application inappropriate to particular circumstances and may constrain regulators' ability to exercise informed judgement. Conversely, excessive discretion permits forbearance driven by political pressure, capture, or risk aversion. Supervisory discretion constitutes an essential feature of effective oversight, enabling regulators to adapt enforcement to institution-specific circumstances and evolving risks, but requires a robust accountability framework to prevent lenience and ensure consistent application.

Case study 15: Horizontal regulation and sectoral boundary challenges

The effectiveness of regulation tends to be challenged where activities transcend traditional sectoral boundaries. BigTech platforms' entry into financial services exemplifies this phenomenon, combining payment services, data processing, advertising, and consumer retail business more generally within integrated business models that fragment across established remits of regulatory authorities. Regulation and supervision are often organised along sectoral lines – with separate rulebooks and distinct authorities for banking, securities, insurance, data protection, competition, and consumer protection. With increasing convergence between these sectors, regulation struggles to address risks arising from business models that span across these sectors.

The "Libra" project, announced in 2019 by Facebook/Meta (later renamed Diem, ultimately abandoned in 2022) illuminates these challenges (Simmons, 2021, p. 83[49]). Libra sought to establish a global digital currency backed by reserves of fiat currencies and short-term government securities, accessible to Meta's 2.7 billion users across Facebook, Instagram, and WhatsApp platforms. The project's scale – potential adoption of the coin might have exceeded that of most national currencies in terms of value – immediately raised concerns regarding monetary sovereignty, financial stability, consumer protection, data privacy, competition, and illicit finance risks.

Relevant regulation was materially and institutionally fragmented, with competences spread across: i) central banks concerned with monetary policy implications; ii) financial regulators responsible for payment systems or for a potential classification of Libra as a financial instrument; iii) competition authorities examining potential abuse of dominant position; iv) data protection authorities applying GDPR-like requirements; and v) anti-money laundering bodies assessing financial crime risks (Simmons, 2021[49]). No single authority possessed a comprehensive mandate spanning all relevant dimensions. The fact that Libra was organised as an association established in Switzerland further complicated questions regarding supervisory primacy, in addition to the known difficulties of cross-jurisdictional co-operation necessary to supervise globally active financial institutions.

The Financial Stability Board and G7 Finance Ministers expressed concerns regarding systemic risk flowing from so-called "stablecoins" when they are issued and used globally, emphasising requirements for adequate governance, comprehensive regulation, and effective supervision (FSB, 2023[27]). The absence of a global and horizontal regulatory structures capable of addressing financial services offered by globally used platforms became apparent.

Meta's ultimate abandonment of the project reflected sustained regulatory opposition, but this opposition manifested primarily through informal pressure and public statements rather than through application of a comprehensive regulatory rulebook fitted to the activity's nature (Simmons, 2021[49]). The episode demonstrated that where regulatory competences fragment across multiple authorities and jurisdictions, even where each authority possesses a mandate regarding particular risk dimensions, collective action problems may impede decisive intervention. Co-ordination costs, divergent priorities, and legal uncertainty regarding the applicability of existing frameworks to novel activities create regulatory gaps.

Relevance for regulatory governance of the digital economy

The challenges regarding institutional effectiveness manifest in financial regulation find direct parallels in regulatory policy for the digital economy, though often with greater complexity given the rapidity of technological change and the need for multidisciplinary expertise.

Institutional capacity and expertise deficits

Regulating digital activities demands specialist knowledge of market practice and technological aspects, spanning, amongst other things, algorithmic or artificial intelligence-driven decision-making, systems design, data analytics, and cybersecurity – expertise not traditionally associated with regulatory authorities (OECD, 2024, pp. 16,18[26]). Hence, the capacity constraints that undermined the effectiveness of financial regulators persist, often more acutely, for the different regulators concerned with regulating the various aspects of the digital economy. The resource requirements for enforcement may substantially exceed those for traditional supervision: regulators should not merely employ technical staff but invest in infrastructure – data analytical platforms, testing environments, forensic tools – representing significant capital and operational costs.

Regulatory perimeter and horizontal integration

Regulation structured along traditional sectoral lines (financial services, telecommunications, media, competition, data protection) struggle to address risks arising from this integration. The perimeter challenges exceed those in financial regulation. Whereas most financial institutions operate primarily within the financial sector (even if spanning banking, securities, and insurance sub-sectors), digital platforms' business models integrate very diverse, including financial, activities to exploit economies of scope and network effects. Each sectoral regulator possesses a mandate covering particular aspects of platform activities but lacks authority over the totality of relevant issues. Clear co-ordination mechanisms are needed as otherwise regulation and supervision remain vulnerable to the same weaknesses exposed by the Financial Crisis, as described (Wilson, Sugimoto and Bains, 2022[34]).

The Wirecard case study is a case in point on how group structures can provoke regulatory ineffectiveness. Digital platforms present similar challenges: a technology company offering payment services may structure operations such that financial activities constitute subsidiaries within a broader corporate group, with the parent entity – controlling data flows, algorithmic systems, and strategic decisions – falling outside financial regulators' direct oversight. Limitations to competence set by the mere structure of legal entities and their relationship structure create opportunities for risks originating in unsupervised entities to propagate into regulated activities. Effective perimeter definition for digital platforms therefore requires functional analysis overcoming corporate structure, with regulatory mandates extending to parent entities and affiliated companies where these control critical operational elements or create material risks to supervised activities (Wilson, Sugimoto and Bains, 2022, pp. 11, 16-18[34]).

Supervisory methodology

Supervision methodology developed for financial institutions, such as risk or outcomes-based regulation (Black, 2005[22]), cannot be easily applied in the context of the digital economy. The digital economy evolves rapidly, rendering historical data limited in predictive value. Enforcement of the rules demands understanding not merely of governance structures but technical implementation. Moreover, digital systems create novel types of risk that do not easily fit traditional risk categories. Algorithmic bias and model drift in machine learning systems, cybersecurity vulnerabilities, data breaches, and system interoperability failures, amongst other things, constitute distinct risk classes requiring specialist skills. Integrating these into overarching risk assessment methodologies demands a substantial effort, and the rapid pace of technological change constantly risks rendering such frameworks obsolete.

The implementation of risk-based approaches becomes particularly challenging given the difficulty in assessing the impact of emerging technologies before widespread adoption. Phenomena based on digital technologies may turn systemic rapidly, leaving regulators insufficient time for prospective risk assessment before a novel market practice becomes critical. The Libra episode demonstrates this difficulty.

Intervention thresholds and discretion

The complexity of setting intervention thresholds and exercising discretion materialises in similar form in the context of regulating the various activities of the digital economy.

First, authorities confronting novel business models face uncertainty regarding intervention thresholds: at what point does a rapidly scaling digital businesses' market penetration warrant pre-emptive supervisory action absent clear evidence of imminent harm? Traditional incentives to delay action – reputational concerns, fear of stifling innovation – intensify where regulated activities involve prominent or "national-champion-type" technology firms with significant economic and political influence. The discretion inherent in principles-based regulatory frameworks, while necessary for technological adaptability, creates scope for forbearance rationalised as regulatory restraint.

Second, detecting non-compliance through traditional supervisory methods becomes problematic. Effective enforcement increasingly requires regulators to employ technological tools – automated monitoring systems, algorithmic auditing capabilities – demanding further specialist capacity development. Lacking this capacity, the factual basis for regulatory intervention is fraught with uncertainty which entails regulatory inaction as a consequence of a prudent approach.

Last, the cross-border operation of digital services complicates enforcement. Fragmented regulatory enforcement across multiple jurisdictions requires a sophisticated design and application of intervention thresholds, and the triggering of supervisory measures typically requires cross-jurisdictional co-ordination and data-sharing.

Regulatory independence and accountability

The balance between regulatory independence and regulatory accountability, which is central to effective regulatory governance, becomes more complex in digital economy contexts, as the information asymmetries between regulated entities and supervisors intensify. Regulatory authorities' dependency upon industry co-operation for technical information and expertise creates structural capture (Ferran, 2012[52]) analogous to those in financial regulation but potentially more acute (in addition to the information asymmetry itself). The multidisciplinary nature of regulating the digital economy multiplies the potential channels through which prior industry relationships might influence regulatory decisions. The Wirecard episode illustrates an additional capture dimension specific to contexts where regulators perceive sectoral reputation as intertwined with national economic interests. Regulation governance for the digital economy faces analogous risks where national technology champions or strategically significant platforms receive different treatment. Institutional cultures and "competitiveness" mandates prioritising sectoral promotion alongside regulatory enforcement generate conflicts requiring explicit resolution through governance structures separating promotional and supervisory functions.

The technical complexity of digital economy regulation compounds these risks: where generalist oversight bodies lack capacity for substantive assessment, institutional capture may prove difficult to detect through traditional accountability channels. Moreover, digital platforms' concentration – a small number of firms dominating particular market segments – magnifies capture risks. Where only a few entities possess deep expertise regarding particular technologies, regulators' dependence upon those entities for information and technical guidance becomes structural rather than incidental.

Lessons for regulatory design and delivery

Specify mandates precisely and establish binding co-ordination mechanisms

The legislative framework establishing regulatory authorities should specify the authority's objectives with precision sufficient to eliminate interpretative ambiguities while providing for conditions for any adaptation to accommodate evolving circumstances (OECD, 2012[4]), Recommendation 7.1. The Mandate should explicitly address co-ordination where multiple authorities share responsibilities and should allocate competences in areas crossing traditional sectoral divisions. Where regulatory objectives require multiple authorities' participation, binding mechanisms – statutory duties to co-operate, joint decision-making requirements, formal information-sharing obligations with enforcement provisions – are essential. There should be gap-filling provisions designating clear responsibility for novel activities.

Ensure resources match regulatory responsibilities

Regulatory effectiveness requires resources proportionate to assigned responsibilities. Digital economy regulation requires comparatively more resources, as it requires multidisciplinary integration of technical, economic, legal, and sectoral expertise. Regulators will need to offer appropriate compensation and retention policies for specialists. Sustainable funding models should recognise that effective digital economy oversight demands continuous and significant infrastructure investment, in particular into real-time data collection, warehousing and processing as well as into the relevant analytical tools.

Implement resource-allocation methodologies

Specific methodologies, such as risk-based or outcomes-based regulation (Black, 2005[22]) can optimise resource allocation when properly designed and implemented on the basis of adequate resources. Supervision of digital platforms should incorporate indicators of rapid scaling potential, establish triggers for intensive review when entities approach systemic significance thresholds, and develop assessment capabilities for novel risk classes.

Co-ordinate regulation across sectoral boundaries

Sectoral regulation proves insufficient for horizontally integrated activities spanning traditional boundaries. Supervisory gaps might emerge where no authority clearly holds mandate, or overlaps may be caused where multiple authorities are competent. Effective regulation requires either comprehensive horizontal rules eliminating sectoral boundaries for specified activities, or enhanced co-ordination mechanisms enabling sectoral regulators to operate coherently, including rules on information sharing, joint decision-making, and the filling of regulatory gaps (OECD, 2012[4]), Principle 1.4. Such whole-of-government approaches, with clearly defined roles and co-ordination structures, are necessary where digital technologies span traditional sectoral boundaries (OECD, 2012[4]; Sarliève et al., 2025[48]).

Prevent regulatory capture

Digital economy regulation faces intensified risk of regulatory capture, as information asymmetries favour regulated entities over supervisors. Where a handful of firms dominate a particular segment, the dependence on those entities for information and guidance increases significantly, transforming the capture risk from incidental into structural. Institutional design should therefore incorporate mechanisms to avoid regulatory capture (OECD, 2012, p. 31[4]).

Institutionalise adaptive capacity

Regulation should incorporate mechanisms for continuous adaptation of material and procedural rules, as technologies, business models, and, accordingly, risks evolve. The regulation of the digital economy demands enhanced adaptive capacity given the speed of technological evolution, as established risk taxonomies become obsolete rapidly, historical precedents provide limited guidance for novel phenomena, and regulatory perimeters require continuous reassessment as platforms integrate across sectors. Adaptive mechanisms should encompass routes for expedited rulemaking in response to emerging risks, cross-jurisdictional co-ordination for rapid information exchange regarding emerging risks and regulatory responses. Agile regulatory governance addresses the design of such adaptive mechanisms, encompassing regulatory experimentation, iterative policy development, and systematic review processes (OECD, 2024[37]; 2025[3]).

7 Regulatory and supervisory co-ordination and alignment

Regulatory co-ordination is indispensable where financial activities extend across jurisdictional boundaries (OECD, 2012[4]), Principle 12.1. The necessity arises from incentive misalignments inherent in jurisdictional fragmentation: regulatory costs concentrate within jurisdictions that implement and enforce regulation, yet these jurisdictions remain affected by risk that propagate across borders from jurisdictions that undercut regulatory standards (Brummer, 2010[55]; Goodhart, 2008[40]). This dynamic incentivises underinvestment in co-ordination mechanisms and generates race-to-bottom pressure where jurisdictions compete for financial sector business.

Therefore, co-ordination failures rather stem from fundamental misalignments in institutional incentives and political economy priorities than from technical obstacles which would be susceptible to procedural solutions. As binding authority remains fragmented across jurisdictions, co-ordination mechanisms can facilitate but not fully substitute for unified supervisory powers. International standard-setting without enforcement mechanisms permits divergence that enables regulatory arbitrage, fragmenting markets along jurisdictional lines while failing to prevent cross-border risk transmission (Wilson, Sugimoto and Bains, 2022[34]).

The following case studies demonstrate three distinct manifestations of co-ordination failure: voluntary information-sharing frameworks that achieve partial improvement but permit persistent measurement inconsistencies; binding operational co-ordination mechanisms that reduce information asymmetries but cannot compel co-operation under stress; and divergent implementation of common standards that enables regulatory arbitrage despite international consensus.

Case study 16: Voluntary information sharing frameworks

Co-ordination mechanisms operate predominantly through soft law – non-binding standards developed by international bodies lacking formal treaty authority (Brummer, 2010[55]). This institutional setup reflects political constraints: jurisdictions retain sovereignty over domestic regulatory frameworks while recognising the necessity of international co-ordination. However, voluntary frameworks and cross-jurisdictional learning and sharing are common practice (Black, 2005[22]) but prove inadequate under crisis conditions when jurisdictions prioritise domestic stakeholder protection over co-ordination, as conflicts over burden-sharing intensify precisely when co-ordination becomes most essential (Turner, 2009, pp. 22-23, 77[14]).

The 2008 crisis exposed a USD 60 trillion global shadow banking sector performing bank-like maturity transformation (entities borrowing on shorter timeframes than they are lending) without co-ordinated oversight (Pozsar et al., 2012[30]). Structured investment vehicles, conduits, and money market funds operated beyond regulatory perimeters. When wholesale funding markets froze, these entities experienced runs without access to central bank liquidity facilities. The collapse of the Reserve Primary Fund following its exposure to the insolvency of Lehman triggered contagion across money market funds holding USD 3.4 trillion, demonstrating the capacity of the shadow banking sector for risk transmission (Gorton and Metrick, 2012[13]).

The G20 Summit in Seoul (2010) mandated the Financial Stability Board to develop comprehensive monitoring addressing definitional fragmentation across jurisdictions. The FSB's October 2011 methodology distinguished activity-based measurement (economic functions) from entity-based measurement (institutional forms). Between 2011 and 2013, FSB working groups negotiated standardised data templates, confidentiality protocols, and reporting obligations amongst 29 jurisdictions representing 80% of global GDP.

Their implementation revealed how complex co-ordination can be. Divergences reflected differing regulatory philosophies rather than technical disagreements. The United States employed broader definitions encompassing money market funds and securities lending; European jurisdictions adopted narrower scopes excluding certain investment fund categories; Asian regulators varied substantially in other respects. All jurisdictions retained discretion over definitions, creating data gaps and inconsistencies, e.g., where entities classified as performing non-bank financial intermediation in one jurisdiction fell outside that category elsewhere.

From 2015 onwards, the FSB's Annual Global Monitoring Reports provided the primary instrument for tracking developments (FSB, 2015-2024[56]). The framework operates through mandatory annual data submissions using standardised templates. While inconsistencies regarding definitions prevent precise cross-jurisdictional risk assessment, the common reporting structure enables regulators to benchmark national developments and identify emerging risk.

The example of the FSB monitoring framework exemplifies that voluntary information-sharing arrangements can establish common infrastructure for regulatory learning but do not lead to consistent implementation where jurisdictions possess differing regulatory philosophies, institutional structures, and approaches to the political economy of the financial markets. Remaining discretion causes measurement gaps, enabling regulatory arbitrage through strategic entity classification and domicile selection.

Case study 17: Reducing information asymmetries by binding co-operation mechanisms

This and similar co-ordination failures are aggravated by information asymmetries. National supervisors possess incomplete data of globally active institutions' operations. Home supervisors lack complete insight into foreign operations while host supervisors possess limited authority over entities headquartered elsewhere (Turner, 2009[14]). Financial institutions exploit these asymmetries, locating high-risk activities in jurisdictions with limited supervisory access to data while maintaining cross-border operations, for example through funding arrangements, common asset exposures, and corporate group structures, that are capable of transmitting the relevant risk across borders. The resulting blind spots prevent individual regulators from assessing system-wide risks.

Prior to the crisis, bilateral memoranda of understanding between supervisors provided ad hoc information exchange on licensing decisions, supervisory methodologies, and material developments. However, these arrangements lacked co-ordination regarding consolidated risk assessment, joint supervisory planning, or crisis management protocols (Avgouleas and Goodhart, 2015[31]). Information-sharing obligations were abandoned when domestic financial stability concerns became paramount. Supervisors discovered hidden exposures in foreign subsidiaries too late; uncoordinated national measures, e.g., ring-fencing of deposit-taking activity, fragmented groups and increased resolution costs; and the absence of burden-sharing mechanisms resulted in negotiations under extreme time pressure, contributing to disorderly failures and the necessity of emergency measures (IMF, 2014[33]; D'Hulster, 2011[57]).

The collapse of Landsbanki in October 2008 illustrates the consequences of these co-ordination failures within the EU single market framework. Operating in the United Kingdom as a branch under EEA passporting rights, Landsbanki raised approximately GBP 4.5 billion in retail deposits while remaining subject to Icelandic home country supervision. When the institution failed, UK depositors found themselves dependent upon the Icelandic deposit insurance scheme and the fiscal resources of a government unable to meet these obligations (Turner, 2009[14]).

The April 2009 G20 London Summit called for enhanced cross-border supervisory co-operation for systemically important institutions. In 2010, the Basel Committee published Good Practice Principles on Supervisory Colleges, establishing a framework for home-host supervisor co-ordination (BCBS, 2010[58]). The framework prescribed the establishment of a college with defined membership based on materiality of cross-border operations, written co-ordination arrangements specifying roles and communication protocols, annual joint risk assessments synthesising supervisory perspectives on group-wide risk profile, co-ordinated supervisory planning to avoid duplication and identify coverage gaps, information-sharing arrangements addressing routine reporting and crisis communication, and crisis preparedness through simulation exercises and pre-agreed escalation procedures.

Between 2011 and 2012, supervisory colleges were established for the 30 initially designated global systemically important financial institutions, typically comprising 8-15 supervisory authorities per institution (FSB, 2014[51]). Core functions included annual joint risk assessments, co-ordinated supervisory plans, crisis simulation exercises, and information-sharing protocols. Membership composition, meeting schedules, and substantive discussions for individual colleges remain confidential supervisory information (BCBS, 2010[58]).

Operational challenges emerged rapidly. Home supervisors possessed superior information through direct oversight but faced criticism from host authorities for insufficient transparency regarding group-wide risk assessments. Confidentiality requirements that existed in relation to certain types of information limited information sharing amongst authorities. In some jurisdictions, the law prohibited disclosure of supervisory information to foreign authorities without explicit authorisation, while concerns about reputational damage created reluctance to share sensitive assessments. This created difficulties where host supervisors required detailed information about group-wide risk concentrations or parent company financial condition to assess implications for local subsidiaries (D'Hulster, 2011[57]).

Binding co-ordination mechanisms reduce information asymmetries during calm market conditions but prove inadequate during a crisis when co-ordination becomes most essential (Avgouleas and Goodhart, 2015[31]). Home-host conflicts over supervisory priorities and burden-sharing intensify precisely when rapid co-ordinated intervention is required (D'Hulster, 2011[57]). Legal constraints on information sharing, resource disparities creating unequal participation, and free-rider dynamics where supervisors benefit from others' disclosures while limiting their own combine to fragment oversight of globally active institutions. The absence of pre-agreed burden-sharing mechanisms results in outcomes determined by bargaining power rather than regulatory principles, incentivising unilateral protective measures that increase aggregate resolution costs (D'Hulster, 2011[57]).

Case study 18: Common standard and divergent implementation

The third manifestation of co-ordination failure concerns divergent implementation of internationally agreed substantive regulatory standards. Where international bodies establish recommendations, jurisdictions retain discretion over domestic implementation, generating inconsistencies that sophisticated market participants exploit through strategic and intentional regulatory structuring. The regulation of stablecoins illustrates this dynamic.

The FSB established recommendations on stablecoins organised around ten functional areas: legal clarity, governance, risk management, data storage and portability, recovery and resolution, addressing risks from stabilisation mechanisms, operational resilience, appropriate settlement finality arrangements, cross-border co-operation, and regulatory scope (FSB, 2023[27]). However, as recommendations, these regulatory mechanisms are non-binding, reflecting the organisation's lack of authority to mandate detailed implementation.

As a consequence, implementation varies from jurisdiction to jurisdiction. The European Union adopted the comprehensive Markets in Crypto-Assets Regulation, restricting issuance of e-money tokens to credit and electronic money institutions, mandating a 100% reserve with at least 60% of reserves for significant tokens held as deposits with credit institutions, and allocating supervisory competences among national authorities and the European Banking Authority (European Union, 2023[59]). MiCAR applies to marketing activities within the EU regardless of issuer domicile.

In the United States, the GENIUS Act establishes a federal framework for payment stablecoins through dual federal-state licensing open to banks, credit union subsidiaries, and approved nonbanks, requiring 100% reserve backing in specified liquid assets, mandating monthly public reserve disclosure, and providing for reciprocity with foreign jurisdictions subject to US-based reserve and OCC registration requirements. Japan extended its Payment Services Act (June 2023), restricting issuance to banks, fund transfer service providers, and licensed trust banks, with reserve requirements varying by issuer category and foreign-issued stablecoins permitted only through domestic intermediaries holding equivalent reserves in Japan. Singapore amended its Payment Services Act 2019 to establish a framework for single-currency stablecoins pegged to the Singapore dollar or G10 currencies, requiring a major payment institution licence above a circulation threshold, 100% backing in highly liquid assets of no more than three-month residual maturity, and initially permitting only domestic issuance (Monetary Authority of Singapore (MAS), 2023[60]).

Divergent approaches to implementing the FSB principles create opportunities for regulatory arbitrage and impose substantial costs on market participants seeking multi-jurisdictional operation. For instance, authorisation requirements vary from the EU's restriction to supervised financial institutions, through the US dual pathway permitting both banks and approved nonbanks, to Japan's limitation to three categories of licensed entities, to Singapore's broader institutional eligibility subject to licensing thresholds. Reserve composition divergences prove particularly consequential: MiCAR's mandatory 60% deposit requirement concentrates counterparty risk, the GENIUS Act's broader permitted assets including repurchase agreements enable greater investment flexibility but appear to be more prone to loss, Japan's issuer-specific requirements create differentiated risk profiles, and Singapore's restriction to highly liquid assets with three-month maximum maturity imposes very conservative standards. Also, cross-border access mechanisms diverge: MiCAR's direct application to foreign issuers marketing in the EU, the GENIUS Act's reciprocity framework dependent upon Treasury determinations combined with extraterritorial reserve requirements, Japan's prohibition of direct foreign issuance requiring domestic intermediation, and Singapore's initial restriction to domestic issuance.

Stablecoin regulation therefore exemplifies that international standard-setting bodies operating through non-binding recommendations cannot achieve effective harmonisation where domestic political economy considerations, institutional competences, and policy priorities provoke divergences. Jurisdictions prioritising innovation adopt a set of rules that follows priorities different from those pursued by jurisdictions emphasising consumer protection or financial stability. These different priorities flow from deeper political economy considerations regarding appropriate balance between market freedom and regulatory intervention, with competing constituencies exerting different influence across jurisdictions and directly influencing the implementation of substantive cross-border standards.

Relevance for regulatory governance of the digital economy

These challenges manifest in similar ways and with particular intensity in the digital economy. While the substantive domains differ, the structural features generating supervisory failures rooted in a lack of co-ordination, resemble one another. The ecosystem of digital platforms, cloud infrastructure providers, artificial intelligence applications, and data intermediaries exhibits characteristics that replicate, and frequently amplify, the co-ordination inefficiencies documented in respect of financial regulation. Four dimensions emerge from the case studies.

First, digital services achieve global market penetration through distribution mechanisms allowing for very speedy market adoption, significantly outpacing the capacity of regulators to respond.

Second, a limited number of critical infrastructure providers dominate digital markets. As a consequence, operational failures propagate sideways across sectors and downward through value chains.

Third, information asymmetries pervade digital economy regulation with greater intensity than in the financial industry. Monitoring globally important infrastructure through a supervisory framework that is fragmented along jurisdictional borders dwarfs the well-known home-host information-sharing difficulties of the financial supervisory colleges (D'Hulster, 2011[57]).

Fourth, divergent implementation of international standards stems from differing policy priorities that contradict substantive harmonisation (Avgouleas and Goodhart, 2015[31]). Prudential authorities were able to converge around shared financial-stability objectives. By contrast, digital economy regulation confronts genuinely conflicting policy goals without obvious hierarchical ordering, reflecting legitimate democratic choices that technical harmonisation cannot resolve.

Contagion channels for risk typically cross jurisdictional borders. At the same time, regulatory fragmentation creates opportunities for regulatory arbitrage and, in the extreme case, may be the start of a development leading to a competitive lowering of regulatory standards. In such a scenario, negative externalities often occur in jurisdictions other than the ones where the relevant risks first were created and where the profits of relevant practices are reaped. The cross-jurisdictional nature of digital technologies necessitates international co-operation mechanisms that promote broad dialogues for sharing evidence and experience, while supporting development of common technical standards to ensure interoperability of emerging technology markets (OECD, 2025[2]; 2024, pp. 10,41[26]). Effectively addressing global challenges requires both national action and international co-operation, with regulation designed to limit unnecessary regulatory divergence while respecting legitimate policy diversity (OECD, 2022[61]). Contemporary governance challenges including overlapping jurisdictions, legal fragmentation, and regulatory obsolescence underscore the imperative for enhanced co-ordination (OECD, 2025[2]).

Lessons for regulatory design and delivery

Design co-ordination mechanisms accounting for non-reciprocal leverage

Host jurisdictions lack powers over foreign-headquartered operators and extraterritorial enforcement mechanisms. Effective supervision therefore requires a conditionality of market access, linking service provision to registration requirements or the establishment of a jurisdictional presence.

Recognise limits of material harmonisation for normative conflicts

Divergent policies reflect legitimate democratic choices regarding appropriate trade-offs between innovation, consumer protection, competition and stability. International standard-setting cannot fully resolve these normative conflicts through material harmonisation. International co-ordination frameworks should acknowledge persistent regulatory diversity, focussing on causes of common concern. Co-operation should focus on improving regulatory effectiveness and addressing transboundary challenges rather than pursuing uniform rules where legitimate policy differences persist (OECD, 2022[61]).

Establish binding home-host protocols before crises materialise

Voluntary co-operation fails under crisis conditions when jurisdictions prioritise domestic stakeholders over foreign creditors or counterparties (FSB, 2023[27]) (D'Hulster, 2011[57]). Emergency arrangements require *ex ante* specification of protocols that establish clear authority allocation, information-sharing obligations, and decision-making procedures under time-constrained conditions.

Establish rapid-response protocols distinct from steady-state co-ordination

Crisis co-ordination demands decision-making that follows set procedures with clear authority allocation, escalation triggers, and communication protocols, which are also appropriate under severe time constraints. The practice followed during routine supervision – consensus-based decision-making, extended consultation periods, voluntary information sharing – prove insufficient when rapid co-ordinated intervention becomes necessary. Fast-track procedures for development and implementation of emergency standards need to be created alongside regular co-ordination frameworks. Periodic joint supervisory exercises should be held for purposes of testing these crisis arrangements.

Reinforce supervisory colleges with real-time data-sharing infrastructure

Periodic information exchange through supervisory colleges proves inadequate for monitoring the globally distributed digital economy. Automated ongoing collection of data on the basis of standardised taxonomies should enable real-time risk assessment across jurisdictions and facilitate identification of systemically significant concentrations. Data-sharing infrastructure requires legal arrangements permitting cross-border information exchange taking into account legitimate confidentiality concerns.

8 Conclusion: Lessons for the transition to the digital economy

The preceding analysis identifies six regulatory governance challenges manifest in both post-crisis financial regulation and contemporary digital economy oversight.

Three challenges concern regulatory capacity and the organisation of information. Section 2 examines information asymmetries between supervisory authorities and regulated entities, showing how inadequate regulatory access to material information enables systemic risk to accumulate beyond supervisory oversight. Section 3 addresses the identification of systemic risk, revealing how established risk taxonomies fail to capture novel contagion mechanisms operating through psychological, operational, and technological channels. Section 4 analyses regulatory adaptability, establishing that effective oversight during fundamental market transformation requires institutional mechanisms enabling both ex ante intervention and functional design of regulatory instruments without comprehensive legislative reform.

Three further challenges concern institutional setup and co-ordination. Section 5 examines path dependencies arising from the timing of intervention, demonstrating how delayed responses permit market structures to entrench and raise the subsequent costs of correction. Section 6 addresses regulatory effectiveness, establishing that mandate calibration, resource adequacy, and intervention thresholds are each required, yet none suffices for supervisory authorities to fulfil their responsibilities. Section 7 analyses international co-ordination, demonstrating that voluntary co-operation, even when institutionalised, proves inadequate for cross-border activities – particularly in crises, when jurisdictions prioritise domestic stakeholder interests.

Each section derives discrete lessons from case study analysis. These lessons, while immediately relevant to digital economy regulatory frameworks, possess broader applicability to regulatory governance during periods of fundamental market transformation. The present synthesis extracts three cross-cutting findings: a) structural parallels between, on the one hand, post-2008 financial regulation and the regulation of Digital Finance, and, on the other hand, the transformation to the digital economy; b) amplification factors that intensify these challenges in digital contexts beyond those characterising financial regulation; and, c) resource implications for societies seeking to establish an effective governance framework for the digital economy.

Table 8.1. Overview of lessons learnt

Challenge	Lesson
Information asymmetries	1. Establish comprehensive reporting infrastructure prior to scaling
	2. Account for cross-border interconnections to avoid regulatory blind spots
	3. Prioritise transparency and reporting to mitigate risks rooted in opacity
Systemic risk	4. Identify critical points of failure across the digital economy to prevent systemic risks
	5. Recognise social contagion in digital markets and the role of algorithmic amplification
	6. Prepare for horizontal and vertical propagation of risk across the digital economy
Regulatory adaptability	7. Anticipatory risk assessment

Challenge	Lesson
	8. Functional reinterpretation of regulatory instruments
	9. Pre-specified emergency frameworks
	10. Periodic perimeter reassessment
Path dependency	11. Foresee regulatory flexibility
	12. Long-term consequences
	13. Reversal of regulatory choices
Regulatory effectiveness	14. Specify mandates precisely and establish binding co-ordination mechanisms
	15. Ensure resources match regulatory responsibilities
	16. Implement resource-allocation methodologies
	17. Co-ordinate regulation across sectoral boundaries
	18. Prevent regulatory capture
	19. Institutionalise adaptive capacity
Co-ordination	20. Design co-ordination mechanisms accounting for non-reciprocal leverage
	21. Recognise limits of material harmonisation for normative conflicts
	22. Establish binding home-host protocols before crises materialise
	23. Establish rapid-response protocols distinct from steady-state co-ordination
	24. Reinforce supervisory colleges with real-time data-sharing infrastructure

Structural parallels between financial regulation and the governance of the digital economy

The regulatory challenges confronting digital economy supervisors substantively replicate those exposed by financial crisis experience and the regulation of Digital Finance. Regulatory inertia – reflecting legislative process constraints, political economy dynamics, or deliberate forbearance – enables risk concentration in unregulated or under-regulated market segments. Information asymmetries favour private actors possessing operational knowledge and technical expertise over supervisory authorities dependent upon reported information.

Delayed regulatory intervention permits early entrants to establish advantages – through network effects, user base accumulation, or technological standard-setting – that subsequently entrench market dominance. Cross-border business and operational structures enable sophisticated entities to exploit jurisdictional fragmentation through regulatory arbitrage, locating activities in favourable regulatory environments while serving global markets. Systemic significance emerges not solely from institutional scale and interconnectedness but from network effects and infrastructure dependencies rendering particular entities critical to market functioning and rendering them single points of failure. In addition, psychological phenomena such as market panic, established as one of the causes of the Financial Crisis, can emerge in digital context faster and at a wider scale.

Amplification of regulatory challenges in digital economy contexts

On the basis of these clear structural parallels, several factors amplify regulatory challenges in digital contexts substantially beyond those characterising financial regulation. Regulators cannot keep pace, as the technological evolution far outruns regulatory adaptation, rendering rules obsolete within short timeframes, and opening gaps between market practice and regulatory coverage. The opacity of algorithmic processes generates information asymmetries different from and greater than those characterising complex financial instruments or off-balance-sheet exposures; supervisory authorities confront systems where algorithmic decision-making processes remain not fully understood even to the entities deploying them, thereby constraining regulatory capacity for substantive assessment. As with finance, supervisory authorities require specialised knowledge predominantly residing within the entities subject to oversight, inverting traditional information flows. Sector-focussed regulators each hold competence over only a part of a cross-sectoral business model, leaving authority fragmented. Digital platforms traverse the competence of specialist sector-based regulators. The provision of remote cross-border services has even fewer enforcement mechanisms characterising financial supervision, where host jurisdictions traditionally possessed authority over local operations. Network effects and the accumulation and combination of data provoke path dependency and lock-in effects within relatively short timeframes as opposed to the decades over which analogous phenomena, such as the securitisation market, or the ascent of online financial offerings, so far developed in financial markets.

Resource implications for effective digital economy governance

Effective regulation demands resource allocation proportionate to assigned responsibilities. This truth applies in particular to infrastructure that allows for real-time monitoring and is capable of supporting the supervision of the digital market at a national and global level. Attracting specialist expertise requires compensation arrangements competitive with technology sector standards. The adoption of a multidisciplinary approach may require costly institutional restructuring.

The regulatory framework necessary for the effective governance of the digital economy demands resource commitments not equivalent, but potentially exceeding those resources required for financial sector regulation. Therefore, failing to invest sufficiently in regulatory frameworks or infrastructure means carrying the risks arising from digital economy.

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